

# Immigration and Credit in America

**J. Anthony Cookson<sup>1</sup>, Benedict Guttman-Kenney<sup>2</sup>, & William Mullins<sup>3</sup>**

<sup>1</sup>University of Colorado Boulder, <sup>2</sup>Rice University, <sup>3</sup>University of California San Diego

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Lone Star Finance Conference, 19 September 2025

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- Immigrants start life in the U.S. *without* a U.S. credit report
  - Any foreign credit history generally unobserved to U.S. lenders

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## What We Do

- Describe **selection** of Immigrants compared to Non-Immigrants in their creditworthiness
- Describe **assimilation** in creditworthiness & access to credit over life cycle
- How **dynamics** of life cycle outcomes vary by one year difference in age of immigration, conditional on birth year & geographic fixed effects

# How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

## Data

- U.S. consumer credit reporting data matched with immigration data

## What We Do

- Describe **selection** of Immigrants & **assimilation** dynamics over life cycle
  - Compare same-age Immigrants vs. Non-Immigrants, Immigrants arriving at different ages

## Immigrants (vs. Non-Immigrants), Immigrants Arriving At Older (vs. Younger) Ages:

- **Positive selection:** More creditworthy (once credit scored)
- **Delayed entry** into credit system
- **Lower long-term access** to auto loans **due to debt aversion**
- Credit card limits are **lower** but become **higher over time**

# Our Paper Contributes To Labor Economics & Household Finance Literatures

- Prior work shows the large positive contribution of legal immigrants to U.S. economy (e.g., Sequeira, Nunn, & Qian, 2020; Kerr & Kerr, 2020; Azoulay, Jones, Kim, & Miranda, 2022; Bernstein, Diamond, McQuade, Jiranaphawiboon, & Pousada, 2025)
- We show the **selection** of immigrants based on their credit behaviors, complementing work focusing on labor market outcomes (e.g., Borjas, 1987; Abramitzsky & Boustan, 2017)
- We show **assimilation** into the U.S. credit markets over life cycle, complementing work on labor & cultural assimilation (e.g., Borjas, 1985; Abramitzsky et al., 2014, 2020, 2021; Lubotsky, 2007; Bleakley & Chin, 2010; Doran, Geber, & Isen, 2022; Bailey et al., 2022)
- We contribute to the household finance literatures on economics of credit information, & credit use across the life cycle that largely has not studied role of immigrants (e.g., Pagano & Jappelli, 1996; Dobbie et al., 2021; Blattner, Hartwig, & Nelson, 2022; Zillessen, 2022; Bach et al., 2023; Ricks & Sandler, 2025; Bakker et al., 2025)



## Roadmap For Presentation

1. Data & Institutional Details
2. Credit Market Entry
3. Creditworthiness
4. Credit Access
5. Mechanisms

## **Data & Institutional Details**

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## University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)

- Anonymized data sample of 1 in 10 consumers with U.S. credit reports
- Monthly data from 2000 to 2024
- Individual tradeline data (e.g., details on each account, including opening dates) & consumer-level data (e.g., birth date, credit score, ZIP code)

Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang  
(2025, *Journal of Economic Literature*)

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## **American Community Survey (ACS), 2010**

- Validate our immigration classification

## **Survey of Consumer Finances (SCF), 2022**

- Evaluate mechanisms behind our results

## Classifying Immigrants In The BTCCP

Before 2012, U.S. Social Security Administration assigned first 5 digits of the 9-digit Social Security Numbers (SSNs) sequentially within states (Puckett, 2009)

Consumers assigned SSNs as adults are very likely to be (legal) immigrants

- validated in prior work (Klopfer & Miller, 2024; Yonker, 2017; Doran, Gelber, & Isen, 2022; Bernstein et al., 2025)
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## Classifying Immigrants In The BTCCP

1. Create lookup table mapping first 5 digits of SSN to SSN assignment year (*SSN Year*)
2. TransUnion merged to their data & returned SSN Year (no SSNs) for each consumer
3. We calculate age of SSN assignment:  $SSN\ Age = SSN\ Year - Birth\ Year$

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**Entrant Sample Of Birth Years 1975 to 1987 With SSN Ages 0 to 29:**

**6,122,932 consumers – 344,261 (5.6%) “Immigrants” (ACS: 6.6% includes illegal)**

Follow **credit outcomes** in BTCCP for **ages 18 to 40**

- Immigrant credit access may be adversely affected by **information frictions**:
  1. **Not having a credit report on arrival**
    - **Lender's prior**: Consumer with no U.S. credit report is **high credit risk**
  2. **After entering the credit system, have a shorter history than Non-Immigrants**
    - Length of credit report has positive impact to credit scores ( $\approx 15\%$  FICO)



## Institutional Details On Lending In America

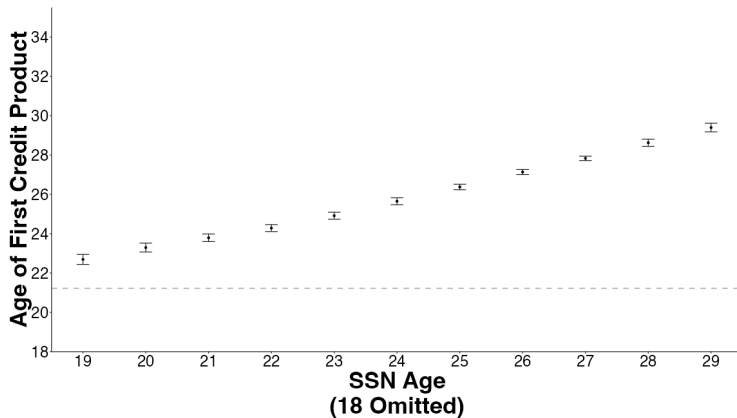
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    - Length of credit report has positive impact to credit scores ( $\approx 15\%$  FICO)
- Different cultural **tastes** & informational barriers for consumers
  - Differences in familiarity & willingness to use debt, understanding of how to build credit score
- Lenders use credit scores (+ other information) in decisions
  - Can use immigration status
  - Cannot use protected characteristics (e.g., gender, national origin, race)  
under Equal Credit Opportunity Act (ECOA)

## Credit Market Entry

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## SSN Assigned At Older Age $\rightarrow$ Later Entry Into U.S. Credit System

$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \epsilon_i \quad (1)$$



# Main Cross-Sectional Regression Specification

Cross-Sectional OLS Regression:

$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \epsilon_i \quad (1)$$

Given **linearity** in *SSN Age*, focus on simpler regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \epsilon_i \quad (2)$$

$\beta_1$ : average difference for Immigrants vs. Non-Immigrants

$\beta_2$ : marginal difference for SSN Age one year later

Cluster standard errors by birth year

Today's results also include fixed effects for geographic controls ( $\eta_{g(i)}$ ):

- First observed ZIP5, Last observed ZIP5, Longest-held ZIP5, & number of ZIP5s

# Immigrants, & Older SSN Age Immigrants, Enter Credit Market Later Than Non-Immigrants & Younger SSN Age Immigrants

$$Y_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (2)$$

| Age at First...          | Credit Report<br>(1) | Credit Product<br>(2) |
|--------------------------|----------------------|-----------------------|
| 21+                      | 1.692***<br>(0.073)  | 1.477***<br>(0.088)   |
| SSN Age                  | 0.727***<br>(0.014)  | 0.689***<br>(0.016)   |
| F.E. Birth Year          | X                    | X                     |
| F.E. Geographic Controls | X                    | X                     |
| $R^2$                    | 0.323                | 0.192                 |
| Mean, SSN Age <21        | 19.86                | 21.25                 |

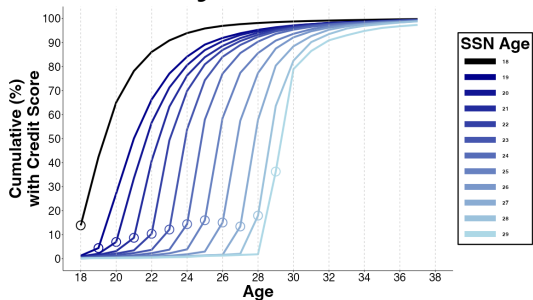
$N = 6,122,932$ , \*\*\* 0.5%, \*\*, 1%, \* 5% statistical significance

# Creditworthiness

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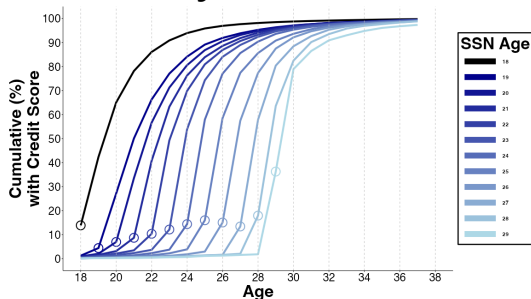
# Once Scored, Immigrants Have Higher Credit Scores Than Non-Immigrants

## A. Any Credit Score

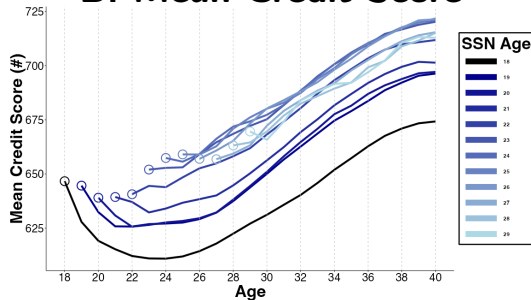


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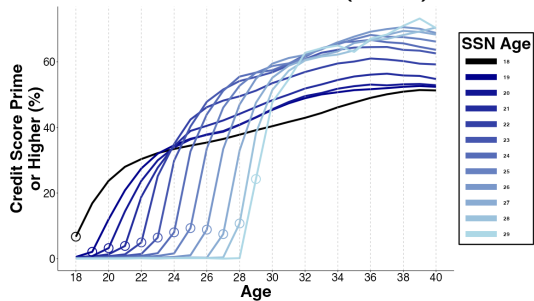
## B. Mean Credit Score



Immigrants (vs. Non-Immigrants) +16 points average scores by age 30, +22 by age 40

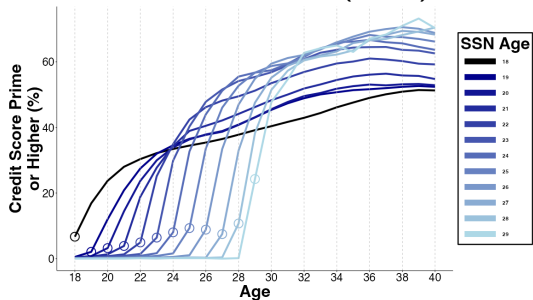


## C. Prime or Higher Credit Score (660+)

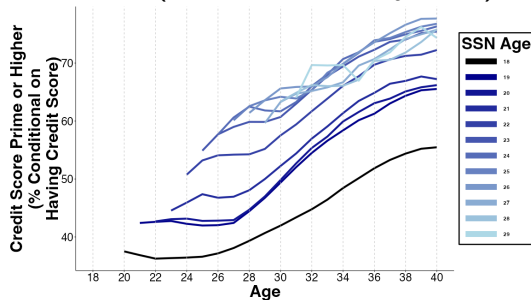


# Positive Selection Of Immigrants In Credit Score Distribution

## C. Prime or Higher Credit Score (660+)



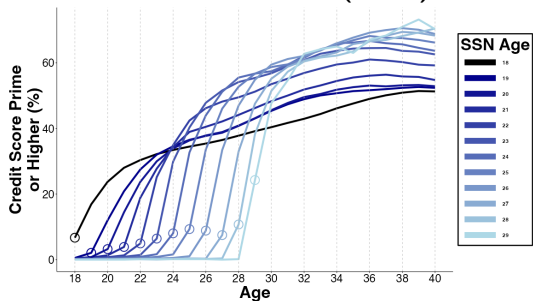
## D. Prime or Higher Credit Score (Conditional On Any Score)



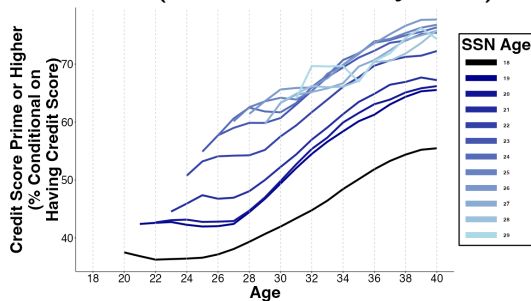
Immigrants:  $\downarrow$  Pr(subprime),  $\uparrow$  Pr(superprime)

# Positive Selection Of Immigrants In Credit Score Distribution

## C. Prime or Higher Credit Score (660+)



## D. Prime or Higher Credit Score (Conditional On Any Score)



Immigrants:  $\downarrow$   $\Pr(\text{subprime})$ ,  $\uparrow$   $\Pr(\text{superprime})$

More creditworthy behaviors:  $\downarrow$  delinquency rates &  $\downarrow$  credit card utilization

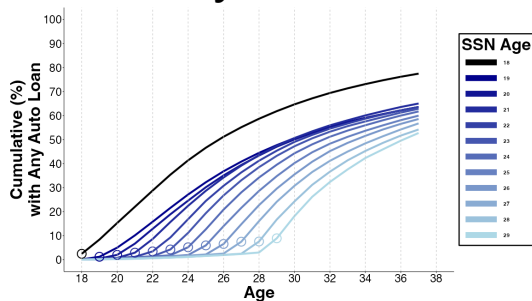
$\downarrow$  5% delinquency rates conditional on credit score

## Credit Access

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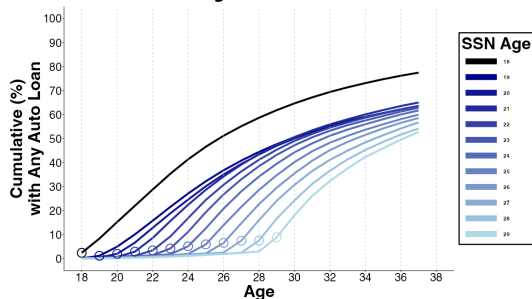
# Immigrants Persistently Less Likely To Hold Auto Loans (& Mortgages)

## A. Any Auto Loan

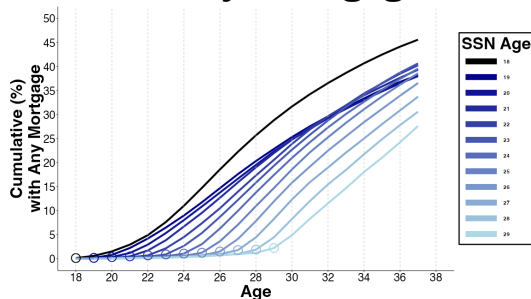


# Immigrants Persistently Less Likely To Hold Auto Loans (& Mortgages)

## A. Any Auto Loan



## B. Any Mortgage



### By Age 37...

- Immigrants are 4.43 (s.e. 0.20) p.p., 6% *less* likely to have auto loan vs. Non-Immigrants
- Each year older SSN Age is 1.06 (0.11) p.p. *less* likely to have auto loan
- Each year older SSN Age is 1.56 (0.09) p.p. *less* likely to have mortgage

# Each Year Later in Life SSN Age Delays First Accessing Mortgage, Auto Loan, & Credit Card By 0.4 to 0.6 Years

$$Y_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (2)$$

| Age at First...          | Auto Loan<br>(1)    | Mortgage<br>(2)     | Credit Card<br>(3)  |
|--------------------------|---------------------|---------------------|---------------------|
| 21+                      | 1.424***<br>(0.136) | -0.082<br>(0.052)   | 1.195***<br>(0.125) |
| SSN Age                  | 0.540***<br>(0.023) | 0.406***<br>(0.017) | 0.644***<br>(0.019) |
| F.E. Birth Year          | X                   | X                   | X                   |
| F.E. Geographic Controls | X                   | X                   | X                   |
| $R^2$                    | 0.169               | 0.154               | 0.159               |
| Mean, SSN Age <21        | 29.15               | 36.36               | 22.68               |

$N = 6,122,932$

## “Paired Cohorts”: Comparing Immigrants vs. One Year Younger SSN Age (Immigrants With Same Birth Year)

$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_{\tau} (\mathbb{I}\{\text{focal cohort}\}_{k(p)} \times \mathbb{I}\{t = \tau\}) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \quad (3)$$

- ‘Focal Cohort’:  $SSN\ Age_c$ , 22-30
- ‘Control Cohort’:  $SSN\ Age_c - 1$ , same birth year as focal. 68 pairs ( $p$ )
- 2,176 cohort-by-year observations ( $k(p), t$ ), s.e. clustered by birth year



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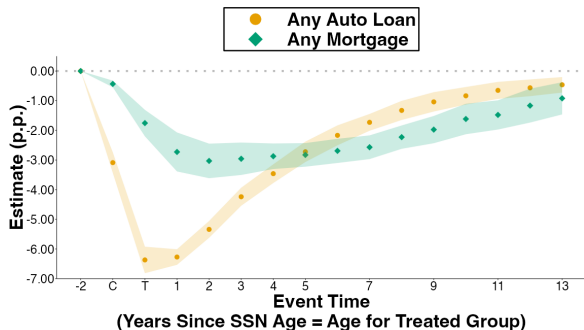
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- 2,176 cohort-by-year observations ( $k(p), t$ ), s.e. clustered by birth year
- **Results:**

**Any Auto Loan after 10 years:**

−0.46 p.p., −0.42 cumulative years

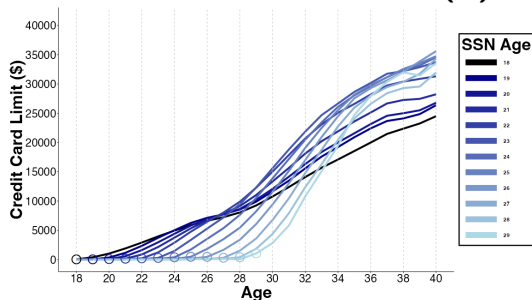
**Any Mortgage after 10 years:**

−0.92 p.p., −0.29 cumulative years



# Immigrants' Credit Card Limits Are Initially Lower Than Non-Immigrants, But Become Higher Than Non-Immigrants Over Life Cycle

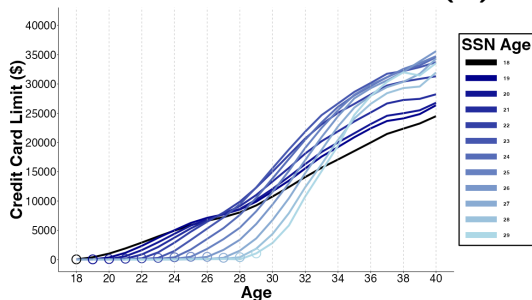
## A. Credit Card Limits (\$)



Similar patterns for number of credit cards,  
credit card limit per card

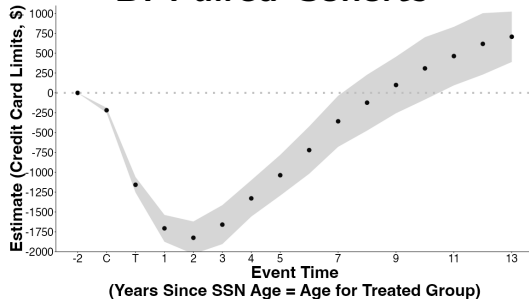
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## A. Credit Card Limits (\$)



Similar patterns for number of credit cards,  
credit card limit per card

## B. Paired Cohorts



**SSN Age +2 Years:**

−\$1,825 (s.e. \$105)

**SSN Age +13 Years:**

+\$707 (s.e. \$162)

# Mechanisms

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# What Mechanisms Explain Less Credit For More Creditworthy Immigrants?

| Potential<br>Mechanisms  | <u>↓ Immigrant (Vs. Non-Immigrant)<br/>Use Of...</u> | <u>↓ Use By Immigrants Arriving<br/>at Older (Vs. Younger) Ages...</u> |           | <u>Immigrants Arriving At Older<br/>(Vs. Younger) Ages Have...</u> |              |
|--------------------------|------------------------------------------------------|------------------------------------------------------------------------|-----------|--------------------------------------------------------------------|--------------|
|                          | Auto Loans                                           | Auto Loans                                                             | Mortgages | Lower Early                                                        | Higher Later |
|                          | (1)                                                  | (2)                                                                    | (3)       | (4)                                                                | (5)          |
| A. Creditworthiness      | ✗                                                    | ✗                                                                      | ✗         | ✗                                                                  | ✓            |
| B. Emigration            | ✗                                                    | ✗                                                                      | ✗         | ✗                                                                  | ✗            |
| C. Tastes                | ✓                                                    | ✓                                                                      | ✓         | ✓                                                                  | ✓            |
| D. Information Frictions |                                                      | ✓                                                                      | ✓         | ✓                                                                  | ✓            |

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|                          |                                              |                                                                |                  | Lower Early                                                | Higher Later              |
|                          | Auto Loans<br>(1)                            | Auto Loans<br>(2)                                              | Mortgages<br>(3) | Credit Card Limits<br>(4)                                  | Credit Card Limits<br>(5) |
| A. Creditworthiness      | ✗                                            | ✗                                                              | ✗                | ✗                                                          | ✓                         |
| B. Emigration            | ✗                                            | ✗                                                              | ✗                | ✗                                                          | ✗                         |
| C. Tastes                | ✓                                            | ✓                                                              | ✓                | ✓                                                          | ✓                         |
| D. Information Frictions |                                              | ✓                                                              | ✓                | ✓                                                          | ✓                         |

## Tastes & Credit Demand:

- Immigrants may be less willing to use debt, or prefer not to purchase cars (or homes)

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| A. Creditworthiness      | ✗                                            | ✗                                                              | ✗                | ✗                                                          | ✓                         |
| B. Emigration            | ✗                                            | ✗                                                              | ✗                | ✗                                                          | ✗                         |
| C. Tastes                | ✓                                            | ✓                                                              | ✓                | ✓                                                          | ✓                         |
| D. Information Frictions |                                              | ✓                                                              | ✓                | ✓                                                          | ✓                         |

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- Immigrants may be less willing to use debt, or prefer not to purchase cars (or homes)
- Immigrants arriving at 1 year older age have –1 year to take out credit (& US exposure)

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|--------------------------|----------------------------------------------|----------------------------------------------------------------|------------------|------------------------------------------------------------|---------------------------|
|                          |                                              |                                                                |                  | Lower Early                                                | Higher Later              |
|                          | Auto Loans<br>(1)                            | Auto Loans<br>(2)                                              | Mortgages<br>(3) | Credit Card Limits<br>(4)                                  | Credit Card Limits<br>(5) |
| A. Creditworthiness      | ✗                                            | ✗                                                              | ✗                | ✗                                                          | ✓                         |
| B. Emigration            | ✗                                            | ✗                                                              | ✗                | ✗                                                          | ✗                         |
| C. Tastes                | ✓                                            | ✓                                                              | ✓                | ✓                                                          | ✓                         |
| D. Information Frictions |                                              | ✓                                                              | ✓                | ✓                                                          | ✓                         |

## Tastes & Credit Demand:

- Immigrants may be less willing to use debt, or prefer not to purchase cars (or homes)
- Immigrants arriving at 1 year older age have –1 year to take out credit (& US exposure)

## Information Frictions:

- Immigrants arrive without a U.S. credit report
- Immigrants have a shorter U.S. credit history



# Immigrants Are Averse To Purchasing Cars With Debt (SCF, 2022)

-22.7 p.p. [95% C.I. -36.7, -8.8], -29% Vs. Non-Immigrants

$$Y_i = \beta_1 \cdot \mathbb{I}\{\text{Immigration Age } 21+\}_i + \beta_2 \cdot \text{Immigration Age}_i + \mu_{b(i)} + \eta_{d(i)} + \epsilon_i$$

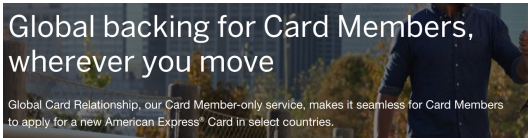
| Dep Var: All Right To Borrow... | <u>In General</u>   | <u>For Vacation</u> | <u>For Living Expenses</u> | <u>For Auto Purchase</u> | <u>For Education</u>  |
|---------------------------------|---------------------|---------------------|----------------------------|--------------------------|-----------------------|
|                                 | (1)                 | (2)                 | (3)                        | (4)                      | (5)                   |
| 21+                             | -0.1499<br>(0.1375) | -0.0774<br>(0.0518) | -0.0804<br>(0.0781)        | -0.2273***<br>(0.0712)   | 0.0905<br>(0.0559)    |
| Immigration Age                 | 0.0100<br>(0.0122)  | 0.0041<br>(0.0056)  | -0.0011<br>(0.0075)        | 0.0113*<br>(0.0060)      | -0.0149**<br>(0.0068) |
| F.E. Birth Year                 | X                   | X                   | X                          | X                        | X                     |
| F.E. Demographic Controls       | X                   | X                   | X                          | X                        | X                     |
| $R^2$                           | 0.030               | 0.018               | 0.029                      | 0.072                    | 0.052                 |
| $N$                             | 1,813               | 1,813               | 1,813                      | 1,813                    | 1,813                 |
| Mean, Immigration Age <21       | 0.0380              | 0.1530              | 0.7338                     | 0.7958                   | 0.8424                |

\*\*\* 1%, \*\* , 5%, \* 10% statistical significance

Immigrants more likely to purchase cars with cash

# Non-Portability Of Credit Histories Across National Boundaries May Prevent Profitable Lending. Recent Market Solutions To Information Friction:

## American Express Global Card Relationship



## Nova Credit

### International credit data

Credit Passport® gives you access to foreign consumer-permissioned credit data that's translated to a local equivalent score to serve the 2 million immigrants that arrive in the United States each year and 280 million immigrants worldwide.

## Equifax Canada Global Consumer Credit File

### Global Consumer Credit File

Helping you grow and retain your clients who are newcomers to Canada

Global Consumer Credit File provides lenders with a seamless and secure platform that connects with global credit bureaus and enables accurate and informed decisioning through robust data.

# Conclusions

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# How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Study **selection & assimilation dynamics** over life cycle of Immigrants vs. Non-Immigrants  
- exploit variation in age of immigration in credit data matched with immigration information

**Immigrants (vs. Non-Immigrants), & Immigrants arriving at older ages:**

- **Positively selected:** more creditworthy (once scored)
- **Delayed entry** into credit system
- **Lower long-term access** to auto loans **due to debt aversion**
- Credit card limits are **lower** but become **higher over time**

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**Implication is that immigrants are profitable to lend to. Potential solutions?**

- **FinTech innovations** (e.g., AmEx Global Card Relationship, NovaCredit, Global Credit File)
- **Removing barriers** to lending to immigrants or sharing information across borders
- **Shorten duration** for retaining positive information on credit reports, but this has trade-offs (e.g., Blattner, Nelson, & Hartwig, 2022; Kovbasyuk & Spagnolo, 2024)

Thank You!

 [www.benedictgk.com](http://www.benedictgk.com)

 [benedictgk@rice.edu](mailto:benedictgk@rice.edu)

 [@gk\\_ben](https://twitter.com/gk_ben)

# Entrant Sample In BTCCP

## Sample Restrictions

- Birth years 1975 - 1987
- SSN Ages 0 to 29 (21 to 29 “Immigrants”)
- Credit report by 2011

## Entrant Sample

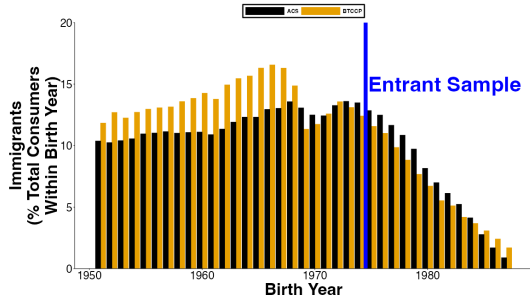
**6,122,932 consumers**

**– 344,261 (5.6%) “Immigrants” (ACS: 6.6% includes illegal)**

**Follow credit outcomes in BTCCP for ages 18 to 40**

## Distribution of Immigrants by...

### A. Birth Year

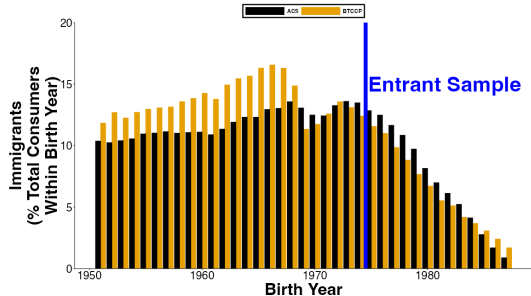




# Immigrant Classification Aligns With American Community Survey (ACS) 🚀

## Distribution of Immigrants by...

### A. Birth Year



### B. Immigration Age



BTCCP: 11.27% SSN Age 21+

ACS: 10.2% Immigrated at Age 21+

Distribution of immigrants aligns by geography. Immigrants more mobile than non-immigrants.

## **“Entrant Sample”: Restrict To Birth Years 1975 - 1987 To Observe U.S. Entry At Ages 21 To 29 & Credit Outcomes For Ages 18 To 40 🍷**

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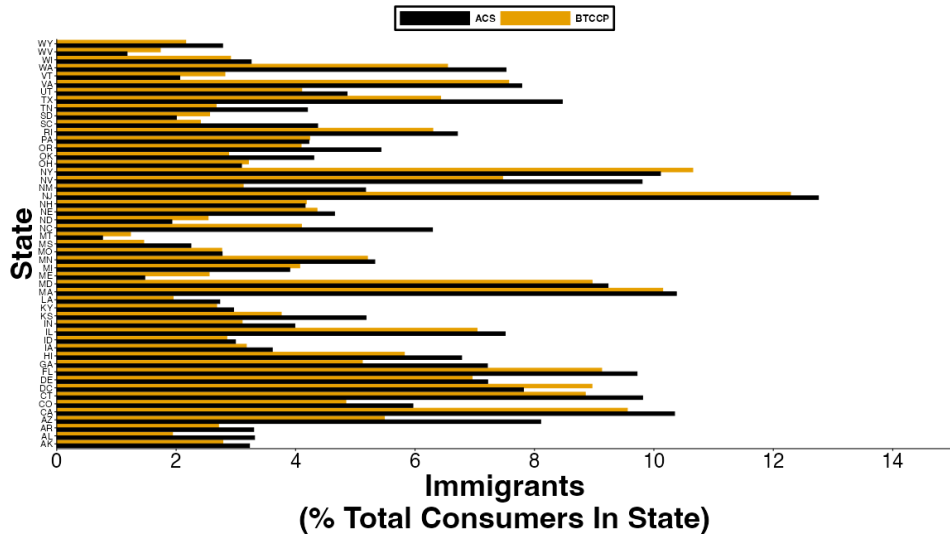
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|                                               |                    |
|-----------------------------------------------|--------------------|
| TransUnion Consumers                          |                    |
| ... born 1951 to 2004                         | 31,869,445         |
| ... with a SSN in TransUnion                  | 25,237,917         |
| <br>Clean Sample                              | <br>18,572,654     |
| ... SSN Age <21                               | 16,478,940         |
| ... SSN Age 21+                               | 2,093,714 (11.27%) |
| ... Immigrant Cohorts (Birth Year x SSN Year) | 775                |
| <br>Entrant Sample (SSN Age < 30)             | <br>6,122,932      |
| ... SSN Age < 21                              | 5,778,671          |
| ... SSN Age 21-29                             | 344,261 (5.62%)    |
| ... Immigrant Cohorts (Birth Year x SSN Year) | 102                |

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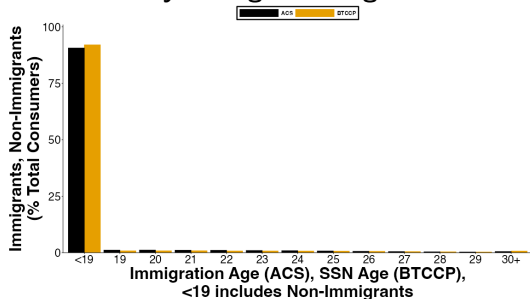
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# Geographic Comparison of Immigrants: ACS vs. BTCCP 🚀

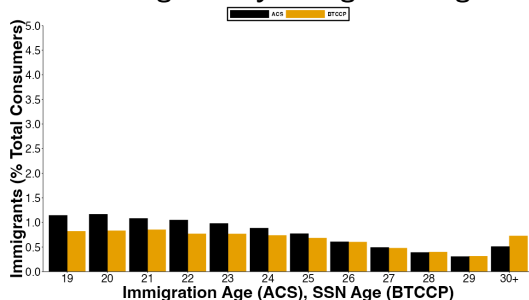


# Immigrant Classification in Entrants Sample: ACS vs. BTCCP

## A. Non-Immigrants and Immigrants by Immigration Age



## B. Immigrants by Immigration Age

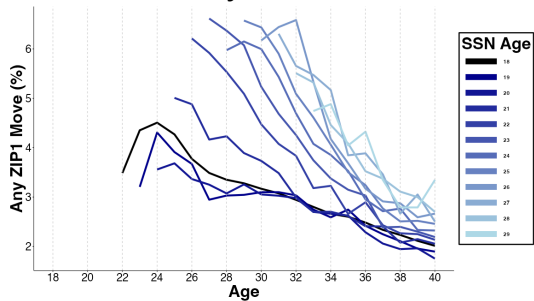


**BTCCP: 5.62% SSN Age 21 to 29**

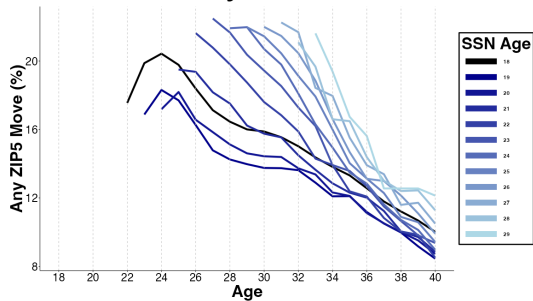
**ACS: 6.57% Immigrated at Age 21 to 29**

# Geographic Mobility 🚀

A. Any ZIP1 Move

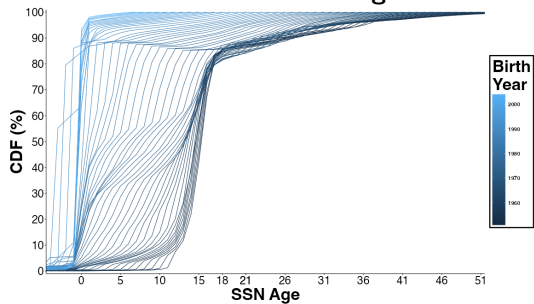


B. Any ZIP5 Move

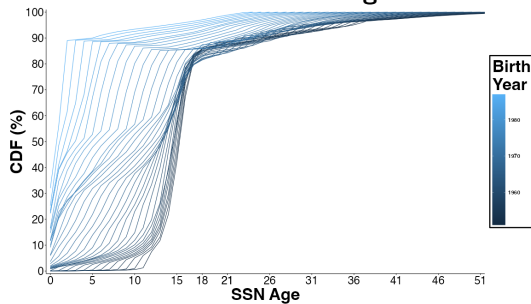


# SSN Ages By Birth Year

**A. Before Cleaning**



**B. After Cleaning**



## Consumers By SSN Age

| SSN Age | Total      | Entrants  |
|---------|------------|-----------|
| <19     | 16,155,157 | 5,677,001 |
| 19      | 175,194    | 50,475    |
| 20      | 148,589    | 51,195    |
| 21      | 148,019    | 52,450    |
| 22      | 141,492    | 47,302    |
| 23      | 145,218    | 47,214    |
| 24      | 149,000    | 45,314    |
| 25      | 147,899    | 42,005    |
| 26      | 142,319    | 36,976    |
| 27      | 133,147    | 29,383    |
| 28      | 123,338    | 24,415    |
| 29      | 113,435    | 19,202    |
| 30+     | 849,847    | 44,642    |

# Sample Frame and Timing of SSN Assignment

| Calendar<br>year | Birth year |      |      |      |      |      |      |      |      |      |      |      |      |
|------------------|------------|------|------|------|------|------|------|------|------|------|------|------|------|
|                  | 1975       | 1976 | 1977 | 1978 | 1979 | 1980 | 1981 | 1982 | 1983 | 1984 | 1985 | 1986 | 1987 |
| 1993             | ≤18        |      |      |      |      |      |      |      |      |      |      |      |      |
| 1994             | 19         | ≤18  |      |      |      |      |      |      |      |      |      |      |      |
| 1995             | 20         | 19   |      |      |      |      |      |      |      |      |      |      |      |
| 1996             | 21         | 20   | ≤18  |      |      |      |      |      |      |      |      |      |      |
| 1997             | 22         | 21   | 20   | 19   | ≤18  |      |      |      |      |      |      |      |      |
| 1998             | 23         | 22   | 21   | 20   | 19   | ≤18  |      |      |      |      |      |      |      |
| 1999             | 24         | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |      |      |      |      |
| 2000             | 25         | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |      |      |      |
| 2001             | 26         | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |      |      |
| 2002             | 27         | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |      |
| 2003             | 28         | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |
| 2004             | 29         | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |
| 2005             | 30         | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |
| 2006             | 31         | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   |
| 2007             | 32         | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   |
| 2008             | 33         | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   |
| 2009             | 34         | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   |
| 2010             | 35         | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   |
| 2011             | 36         | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   |
| 2012             | 37         | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   |
| 2013             | 38         | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   |
| 2014             | 39         | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   |
| 2015             | 40         | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   |
| 2016             | 41         | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   |
| 2017             | 42         | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   |
| 2018             | 43         | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   |
| 2019             | 44         | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   |
| 2020             | 45         | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   |
| 2021             | 46         | 45   | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   |
| 2022             | 47         | 46   | 45   | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   |
| 2023             | 48         | 47   | 46   | 45   | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   |
| 2024             | 49         | 48   | 47   | 46   | 45   | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   |

Age at SSN  
assignment

Years of  
data in our  
sample

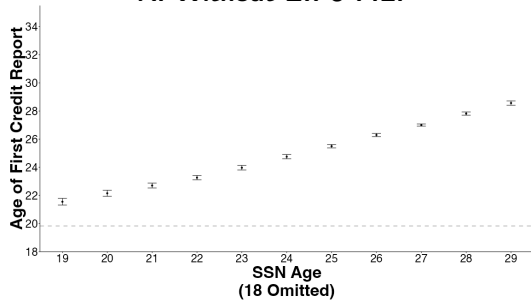


# Summary Statistics

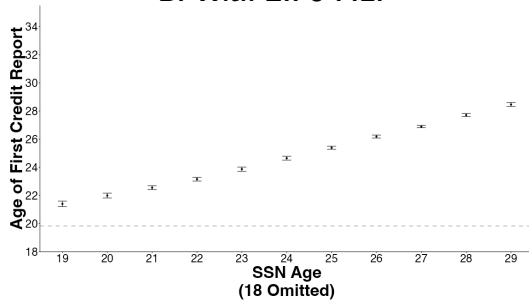
|                            | Non-Immigrants<br>SSN Age $\geq 21$ |             | Immigrants<br>SSN Age $\geq 21$ |             |
|----------------------------|-------------------------------------|-------------|---------------------------------|-------------|
|                            | Mean                                | # Consumers | Mean                            | # Consumers |
| <b>Credit Report</b>       |                                     |             |                                 |             |
| Age at First Credit Report | 19.86                               | 5,778,671   | 25.21                           | 344,261     |
| <b>Credit Score</b>        |                                     |             |                                 |             |
| Any Score by Age 30 (%)    | 97.57                               | 5,778,671   | 88.14                           | 344,261     |
| Credit Score at Age 30     | 626.1                               | 5,461,785   | 660.7                           | 293,349     |
| Any Score by Age 40 (%)    | 99.61                               | 4,487,804   | 99.08                           | 312,391     |
| Credit Score at Age 40     | 659.5                               | 4,179,752   | 698.9                           | 270,177     |
| <b>Auto Loan</b>           |                                     |             |                                 |             |
| Any Auto Loan (%)          | 82.11                               | 5,778,671   | 69.07                           | 344,261     |
| Age at First Auto Loan     | 26.03                               | 4,745,108   | 30.30                           | 237,797     |
| <b>Mortgage</b>            |                                     |             |                                 |             |
| Any Mortgage (%)           | 51.13                               | 5,778,671   | 47.62                           | 344,261     |
| Age at First Mortgage      | 29.49                               | 2,954,699   | 32.73                           | 163,937     |
| <b>Credit Card</b>         |                                     |             |                                 |             |
| Any Credit Card (%)        | 97.16                               | 5,778,671   | 97.85                           | 344,261     |
| Age at First Credit Card   | 22.07                               | 5,614,625   | 26.42                           | 336,869     |
| <b>Credit Card Limits</b>  |                                     |             |                                 |             |
| Age 30                     | \$11,733                            | 5,778,671   | \$11,193                        | 344,261     |
| Age 40                     | \$22,477                            | 4,487,804   | \$28,158                        | 312,391     |

# Age At First Credit Report

**A. Without ZIP5 F.E.**

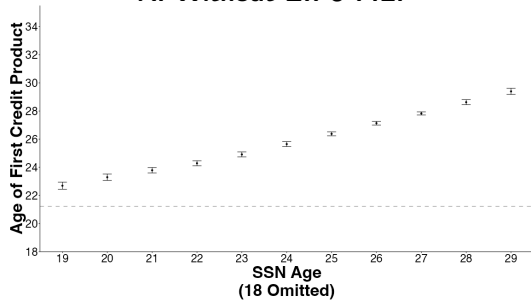


**B. With ZIP5 F.E.**

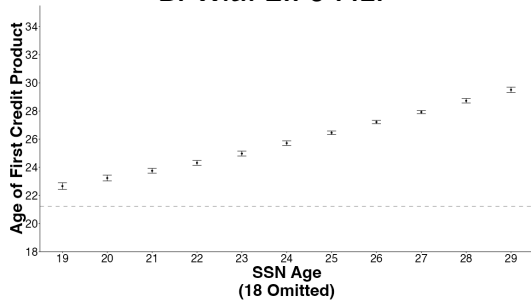


# Age At First Credit Product

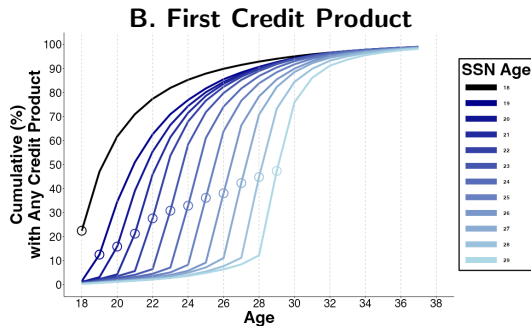
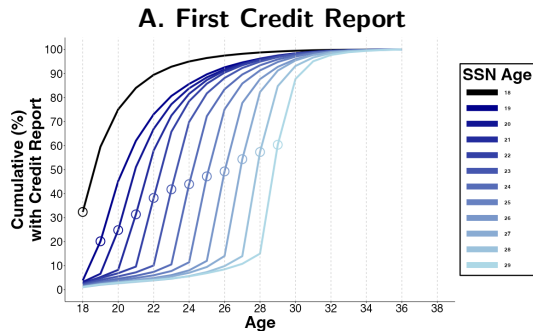
**A. Without ZIP5 F.E.**



**B. With ZIP5 F.E.**



# Age At First Credit Report and First Credit Product By SSN Age Cohort



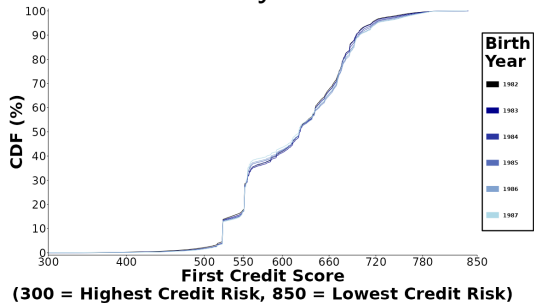
| Dep Var: Age at First... | <u>Credit Report</u> |                     |                     | <u>Credit Product</u> |                     |                     |
|--------------------------|----------------------|---------------------|---------------------|-----------------------|---------------------|---------------------|
|                          | (1)                  | (2)                 | (3)                 | (4)                   | (5)                 | (6)                 |
| 21+                      | 1.989***<br>(0.119)  | 1.972***<br>(0.101) | 1.809***<br>(0.088) | 1.680***<br>(0.129)   | 1.665***<br>(0.112) | 1.618***<br>(0.106) |
| SSN Age                  | 0.782***<br>(0.027)  | 0.740***<br>(0.015) | 0.744***<br>(0.014) | 0.744***<br>(0.026)   | 0.704***<br>(0.017) | 0.720***<br>(0.016) |
| Birth Year F.E.          |                      | X                   | X                   |                       | X                   | X                   |
| First Zip5 F.E.          |                      |                     | X                   |                       |                     | X                   |
| $R^2$                    | 0.239                | 0.267               | 0.289               | 0.077                 | 0.087               | 0.129               |
| $N$                      | 6,122,932            | 6,122,932           | 6,122,932           | 6,122,932             | 6,122,932           | 6,122,932           |
| Mean, SSN Age <21        | 19.864               | 19.864              | 19.864              | 21.251                | 21.251              | 21.251              |

## Credit Entry With Additional F.E.

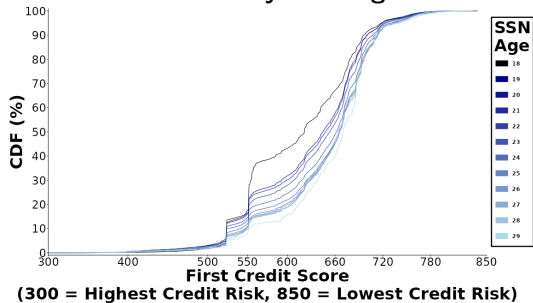
| Dep Var: Age at First... | <u>Credit Report</u>  |                       |                       | <u>Credit Product</u> |                       |                       |
|--------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                          | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   | (6)                   |
| 21+                      | 1.7757***<br>(0.0787) | 1.7721***<br>(0.0779) | 1.6915***<br>(0.0733) | 1.6322***<br>(0.0907) | 1.6353***<br>(0.0884) | 1.477***<br>(0.0878)  |
| SSN Age                  | 0.7466***<br>(0.0133) | 0.7463***<br>(0.0133) | 0.7273***<br>(0.0144) | 0.7271***<br>(0.0155) | 0.7264***<br>(0.0154) | 0.6886***<br>(0.0157) |
| Birth Year F.E.          | X                     | X                     | X                     | X                     | X                     | X                     |
| First Zip5 F.E.          | X                     | X                     | X                     | X                     | X                     | X                     |
| Last Zip5 F.E.           | X                     | X                     | X                     | X                     | X                     | X                     |
| Longest Zip5 F.E.        |                       | X                     | X                     |                       | X                     | X                     |
| Number Zip5 F.E.         |                       |                       | X                     |                       |                       | X                     |
| $R^2$                    | 0.299                 | 0.299                 | 0.323                 | 0.152                 | 0.156                 | 0.192                 |
| $N$                      | 6,122,932             | 6,122,932             | 6,122,932             | 6,122,932             | 6,122,932             | 6,122,932             |
| Mean, SSN Age <21        | 19.864                | 19.864                | 19.864                | 21.2508               | 21.2508               | 21.2508               |

# CDF Of First Credit Score 🚀

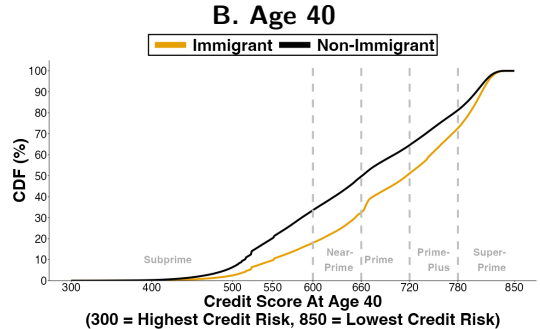
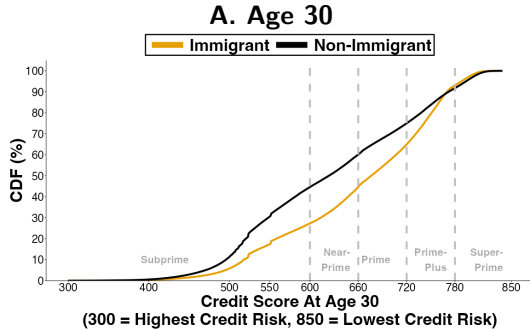
## A. CDF By Birth Year



## B. CDF By SSN Age

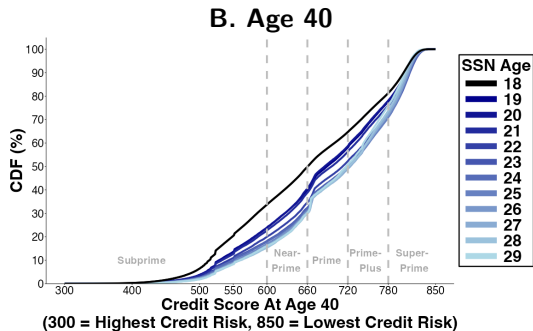
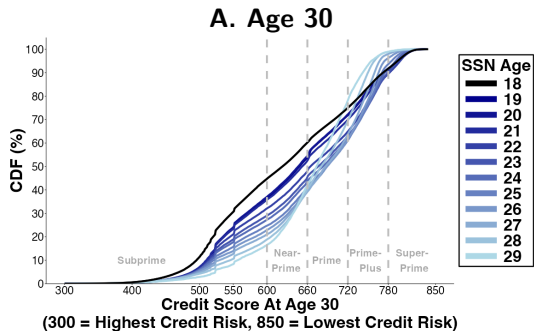


# CDF Of Credit Score At Ages 30 & 40, By Immigration Status 🇺🇸

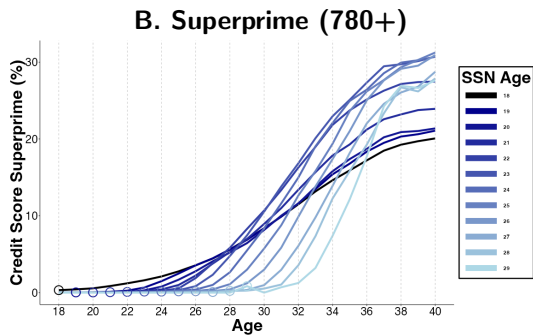
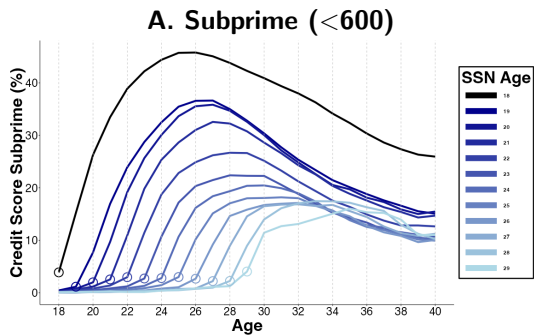




# CDF Of Credit Score At Ages 30 & 40, By SSN Age 🗺️



# Lifecycle Of The Distribution Of Credit Scores By SSN Age Cohorts 🚀



# Immigrant Credit Scores Higher In Spite Of Shorter Credit Histories

|                          | Credit Score at |           | Any Prime or Higher<br>(660+) Credit Score at |           |
|--------------------------|-----------------|-----------|-----------------------------------------------|-----------|
|                          | Age 30          | Age 40    | Age 30                                        | Age 40    |
|                          | (1)             | (2)       | (3)                                           | (4)       |
| 21+                      | 15.7***         | 21.6***   | 6.6***                                        | 2.0***    |
|                          | (1.2)           | (0.6)     | (0.6)                                         | (0.2)     |
| SSN Age                  | 1.6***          | 1.5***    | -0.1                                          | 1.6***    |
|                          | (0.3)           | (0.2)     | (0.2)                                         | (0.1)     |
| F.E. Birth Year          | X               | X         | X                                             | X         |
| F.E. Geographic Controls | X               | X         | X                                             | X         |
| $R^2$                    | 0.202           | 0.200     | 0.156                                         | 0.152     |
| $N$                      | 5,755,134       | 4,449,929 | 6,122,932                                     | 4,800,195 |
| Mean, SSN Age <21        | 626.1           | 659.5     | 37.71                                         | 46.84     |

# Credit Scores At Ages 30 & 40, 🗺️

| Dep Var: Credit Score at ... |                  | Age 30           |                  |                  | Age 40           |                  |
|------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                              | (1)              | (2)              | (3)              | (4)              | (5)              | (6)              |
| 21+                          | 23.3***<br>(2.3) | 23.7***<br>(2.1) | 18.8***<br>(1.6) | 31.5***<br>(0.6) | 32.3***<br>(0.7) | 25.8***<br>(0.7) |
| SSN Age                      | 2.7***<br>(0.5)  | 3.2***<br>(0.4)  | 2.0***<br>(0.4)  | 1.7***<br>(0.4)  | 2.4***<br>(0.2)  | 1.5***<br>(0.2)  |
| Birth Year F.E.              |                  | X                | X                |                  | X                | X                |
| First Zip5 F.E.              |                  |                  | X                |                  |                  | X                |
| $R^2$                        | 0.005            | 0.009            | 0.135            | 0.008            | 0.022            | 0.132            |
| $N$                          | 5,755,134        | 5,755,134        | 5,755,134        | 4,449,929        | 4,449,929        | 4,449,929        |
| Mean, SSN Age <21            | 626.1            | 626.1            | 626.1            | 659.5            | 659.5            | 659.5            |

# Credit Scores At Ages 30 & 40, With Additional F.E. 📌

| Dep Var: Credit Score at... |                  | Age 30           |                  |                  | Age 40           |                  |
|-----------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                             | (1)              | (2)              | (3)              | (4)              | (5)              | (6)              |
| 21+                         | 15.7***<br>(1.3) | 15.4***<br>(1.2) | 15.7***<br>(1.2) | 21.6***<br>(0.6) | 21.1***<br>(0.6) | 21.6***<br>(0.6) |
| SSN Age                     | 1.5***<br>(0.3)  | 1.5***<br>(0.3)  | 1.6***<br>(0.3)  | 1.3***<br>(0.1)  | 1.3***<br>(0.1)  | 1.5***<br>(0.2)  |
| Birth Year F.E.             | X                | X                | X                | X                | X                | X                |
| First Zip5 F.E.             | X                | X                | X                | X                | X                | X                |
| Last Zip5 F.E.              | X                | X                | X                | X                | X                | X                |
| Longest Zip5 F.E.           |                  | X                | X                |                  | X                | X                |
| Number Zip5 F.E.            |                  |                  | X                |                  |                  | X                |
| $R^2$                       | 0.193            | 0.201            | 0.202            | 0.192            | 0.199            | 0.200            |
| $N$                         | 5,755,134        | 5,755,134        | 5,755,134        | 4,449,929        | 4,449,929        | 4,449,929        |
| Mean, SSN Age <21           | 626.1            | 626.1            | 626.1            | 659.5            | 659.5            | 659.5            |

## Credit Score Prime or Higher (660+) At Ages 30 & 40, 🍷

| Dep Var: Prime or Higher at ... |                   | Age 30            |                   |                   | Age 40            |                   |
|---------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                 | (1)               | (2)               | (3)               | (4)               | (5)               | (6)               |
| 21+                             | 9.35***<br>(0.93) | 9.50***<br>(0.83) | 7.63***<br>(0.74) | 3.86***<br>(0.31) | 4.08***<br>(0.29) | 2.01***<br>(0.14) |
| SSN Age                         | 0.01<br>(0.28)    | 0.29<br>(0.23)    | -0.08<br>(0.20)   | 1.71***<br>(0.14) | 1.92***<br>(0.10) | 1.58***<br>(0.09) |
| Birth Year F.E.                 |                   | X                 | X                 |                   | X                 | X                 |
| First Zip5 F.E.                 |                   |                   | X                 |                   |                   | X                 |
| $R^2$                           | 0.002             | 0.006             | 0.104             | 0.004             | 0.010             | 0.092             |
| $N$                             | 6,122,932         | 6,122,932         | 6,122,932         | 4,800,195         | 4,800,195         | 4,800,195         |
| Mean, SSN Age <21               | 37.71             | 37.71             | 37.71             | 46.84             | 46.84             | 46.84             |

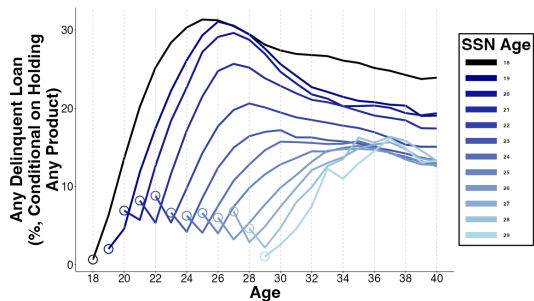
# Credit Score Prime or Higher (660+) At Ages 30 & 40, With Additional F.E.



| Dep Var: Prime or Higher Score at... | Age 30            |                   |                   | Age 40            |                   |                   |
|--------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                      | (1)               | (2)               | (3)               | (4)               | (5)               | (6)               |
| 21+                                  | 6.45***<br>(0.60) | 6.32***<br>(0.55) | 6.60***<br>(0.57) | 1.32***<br>(0.23) | 1.19***<br>(0.27) | 1.99***<br>(0.21) |
| SSN Age                              | -0.22<br>(0.18)   | -0.22<br>(0.18)   | -0.14<br>(0.19)   | 1.43***<br>(0.07) | 1.43***<br>(0.07) | 1.58***<br>(0.09) |
| Birth Year F.E.                      | X                 | X                 | X                 | X                 | X                 | X                 |
| First Zip5 F.E.                      | X                 | X                 | X                 | X                 | X                 | X                 |
| Last Zip5 F.E.                       | X                 | X                 | X                 | X                 | X                 | X                 |
| Longest Zip5 F.E.                    |                   | X                 | X                 |                   | X                 | X                 |
| Number Zip5 F.E.                     |                   |                   | X                 |                   |                   | X                 |
| $R^2$                                | 0.149             | 0.155             | 0.156             | 0.141             | 0.146             | 0.152             |
| $N$                                  | 6,122,932         | 6,122,932         | 6,122,932         | 4,800,195         | 4,800,195         | 4,800,195         |
| Mean, SSN Age <21                    | 37.71             | 37.71             | 37.71             | 46.84             | 46.84             | 46.84             |

# Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

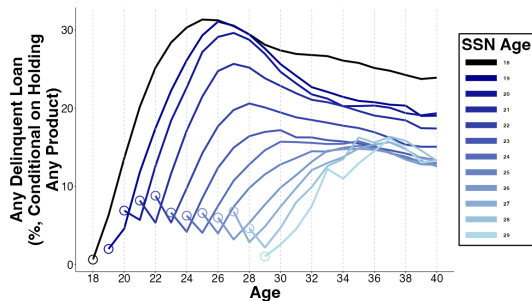
## A. Any Delinquency



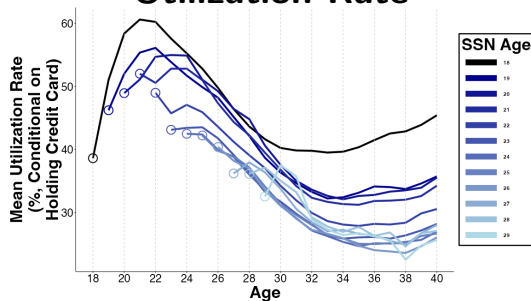


# Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

## A. Any Delinquency

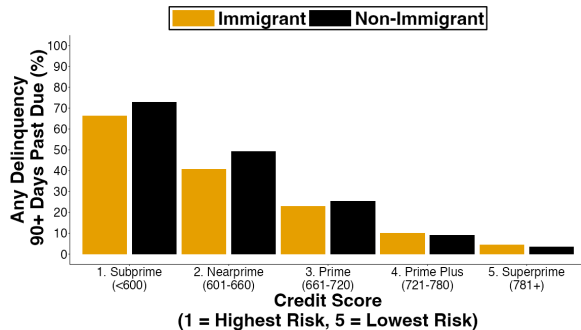


## B. Credit Card Utilization Rate



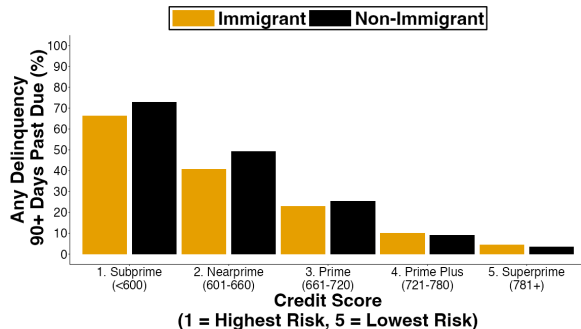
↓ 5% delinquency rates at age 37 conditional on credit score at age 30

# Immigrants (vs. Non-Immigrants) Are More Creditworthy: 5% Lower Delinquency Rates At Age 37 Conditional On Age 30 Credit Score



# Immigrants (vs. Non-Immigrants) Are More Creditworthy:

## 5% Lower Delinquency Rates At Age 37 Conditional On Age 30 Credit Score



|                          |                        |
|--------------------------|------------------------|
| 21+                      | -0.0140***<br>(0.0041) |
| SSN Age                  | -0.0004<br>(0.0005)    |
| F.E. Birth Year          | X                      |
| F.E. Age 30 Zip5         | X                      |
| F.E. Age 30 Credit Score | X                      |
| $R^2$                    | 0.185                  |
| $N$                      | 5,755,134              |
| Mean, SSN Age <21        | 0.2722                 |

# Calibration Bias: Delinquency At 37 Conditional On Age 30 Credit Score

| Dep Var: Any Delinquency<br>by age 37 | <u>All Credit Scores</u><br>(1) | <u>Subprime</u><br>(2) | <u>Nearprime</u><br>(3) | <u>Prime</u><br>(4)    | <u>Prime Plus</u><br>(5) | <u>Superprime</u><br>(6) |
|---------------------------------------|---------------------------------|------------------------|-------------------------|------------------------|--------------------------|--------------------------|
| 21+                                   | -0.0140***<br>(0.0041)          | -0.0369***<br>(0.0074) | -0.0272***<br>(0.0058)  | 0.0006<br>(0.0049)     | 0.0099***<br>(0.0019)    | 0.0037<br>(0.0028)       |
| SSN Age                               | -0.0004<br>(0.0005)             | 0.0023<br>(0.0012)     | -0.0019<br>(0.0009)     | -0.0034***<br>(0.0007) | -0.0021***<br>(0.0003)   | -0.0003<br>(0.0007)      |
| F.E. Birth Year                       | X                               | X                      | X                       | X                      | X                        | X                        |
| F.E. Age 30 Zip5                      | X                               | X                      | X                       | X                      | X                        | X                        |
| F.E. Age 30 Credit Score              | X                               | X                      | X                       | X                      | X                        | X                        |
| $R^2$                                 | 0.185                           | 0.032                  | 0.028                   | 0.027                  | 0.025                    | 0.022                    |
| $N$                                   | 5,755,134                       | 2,515,933              | 898,118                 | 871,578                | 999,130                  | 470,375                  |
| Mean, SSN Age <21                     | 0.2722                          | 0.4553                 | 0.2620                  | 0.1314                 | 0.0451                   | 0.0163                   |

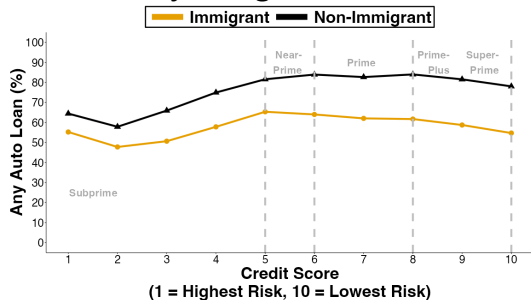


# Age 37 Credit Outcomes Conditional On Age 30 Credit Score 📊

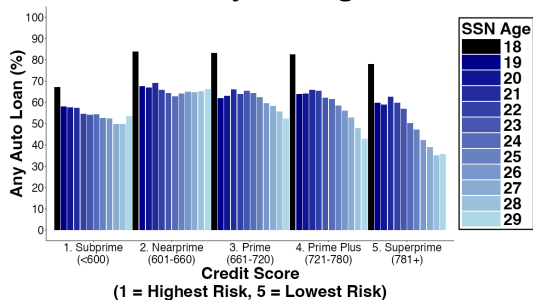
| Dep Var: Outcomes at Age 37... | <u>Any Auto Loan</u>   | <u>Any Mortgage</u>    | <u>Any Credit Card</u> | <u>Number of Credit Cards</u> | <u>Credit Card Limits</u> |
|--------------------------------|------------------------|------------------------|------------------------|-------------------------------|---------------------------|
|                                | (1)                    | (2)                    | (3)                    | (4)                           | (5)                       |
| 21+                            | -0.0889***<br>(0.0039) | -0.0233***<br>(0.0050) | -0.0161***<br>(0.0013) | 0.2174***<br>(0.0324)         | 981.1<br>(561.8)          |
| SSN Age                        | -0.0142***<br>(0.0006) | -0.0235***<br>(0.0012) | 0.0005<br>(0.0003)     | 0.0435***<br>(0.0110)         | -434.6**<br>(131.7)       |
| F.E. Birth Year                | X                      | X                      | X                      | X                             | X                         |
| F.E. Age 30 Zip5               | X                      | X                      | X                      | X                             | X                         |
| F.E. Age 30 Credit Score       | X                      | X                      | X                      | X                             | X                         |
| $R^2$                          | 0.108                  | 0.268                  | 0.110                  | 0.142                         | 0.306                     |
| $N$                            | 5,755,134              | 5,755,134              | 5,755,134              | 5,755,134                     | 5,755,134                 |
| Mean, SSN Age <21              | 0.7534                 | 0.4473                 | 0.9221                 | 3.9179                        | 20515.6                   |

# Any Auto Loan By Age 37, Conditional On Age 30 Credit Score 🚀

## A. By Immigration Status



## B. By SSN Age

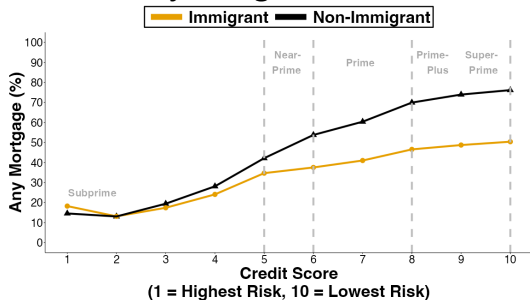




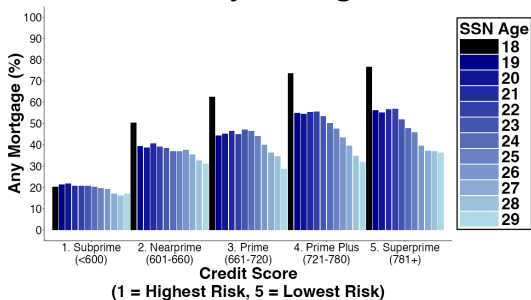


# Any Mortgage By Age 37, Conditional On Age 30 Credit Score 🚀

## A. By Immigration Status

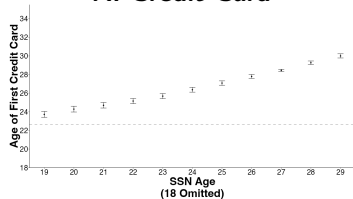


## B. By SSN Age

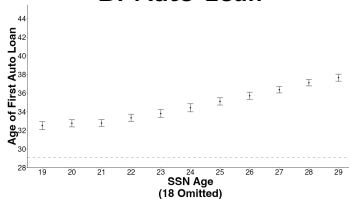


# Age At First Credit Product By Type 🚀

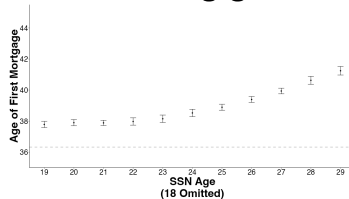
## A. Credit Card



## B. Auto Loan



## C. Mortgage



| Dep Var: Age at First... | <u>Credit Card</u>  |                     |                     | <u>Auto Loan</u>    |                     |                     | <u>Mortgage</u>     |                     |                     |
|--------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                          | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 | (9)                 |
| 21+                      | 1.172***<br>(0.182) | 1.160***<br>(0.169) | 1.367***<br>(0.158) | 2.954***<br>(0.239) | 2.930***<br>(0.217) | 1.967***<br>(0.217) | 0.831***<br>(0.124) | 0.781***<br>(0.097) | 0.219*<br>(0.083)   |
| SSN Age                  | 0.696***<br>(0.030) | 0.663***<br>(0.022) | 0.691***<br>(0.020) | 0.673***<br>(0.045) | 0.612***<br>(0.023) | 0.619***<br>(0.024) | 0.504***<br>(0.048) | 0.403***<br>(0.018) | 0.432***<br>(0.018) |
| Birth Year F.E.          |                     | X                   | X                   |                     | X                   | X                   |                     | X                   | X                   |
| First Zip5 F.E.          |                     |                     | X                   |                     |                     | X                   |                     |                     | X                   |
| $R^2$                    | 0.029               | 0.033               | 0.085               | 0.025               | 0.030               | 0.082               | 0.008               | 0.024               | 0.081               |
| $N$                      | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           |
| Mean, SSN Age <21        | 22.677              | 22.677              | 22.677              | 29.151              | 29.151              | 29.151              | 36.356              | 36.356              | 36.356              |

# Credit Access With Additional F.E.

| Dep Var: Age at First... | <u>Credit Card</u>    |                       |                       |                       | <u>Auto Loan</u>      |                       |                       | <u>Mortgage</u>       |                       |
|--------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                          | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   | (6)                   | (7)                   | (8)                   | (9)                   |
| 21+                      | 1.4213***<br>(0.1344) | 1.4318***<br>(0.1308) | 1.1952***<br>(0.1250) | 1.7944***<br>(0.1736) | 1.7860***<br>(0.1729) | 1.4242***<br>(0.1356) | 0.0911<br>(0.0653)    | 0.0735<br>(0.0671)    | -0.0822<br>(0.0518)   |
| SSN Age                  | 0.7010***<br>(0.0191) | 0.7000***<br>(0.0190) | 0.6435***<br>(0.0194) | 0.6271***<br>(0.0208) | 0.6255***<br>(0.0215) | 0.5401***<br>(0.0230) | 0.4454***<br>(0.0162) | 0.4434***<br>(0.0163) | 0.4057***<br>(0.0173) |
| Birth Year F.E.          | X                     | X                     | X                     | X                     | X                     | X                     | X                     | X                     | X                     |
| First Zip5 F.E.          | X                     | X                     | X                     | X                     | X                     | X                     | X                     | X                     | X                     |
| Last Zip5 F.E.           | X                     | X                     | X                     | X                     | X                     | X                     | X                     | X                     | X                     |
| Longest Zip5 F.E.        |                       | X                     | X                     |                       | X                     | X                     |                       | X                     | X                     |
| Number Zip5 F.E.         |                       |                       | X                     |                       |                       | X                     |                       |                       | X                     |
| $R^2$                    | 0.114                 | 0.118                 | 0.159                 | 0.120                 | 0.125                 | 0.169                 | 0.131                 | 0.143                 | 0.154                 |
| $N$                      | 6,122,932             | 6,122,932             | 6,122,932             | 6,122,932             | 6,122,932             | 6,122,932             | 6,122,932             | 6,122,932             | 6,122,932             |
| Mean, SSN Age <21        | 22.6771               | 22.6771               | 22.6771               | 29.1511               | 29.1511               | 29.1511               | 36.3559               | 36.3559               | 36.3559               |

# Credit Access By Age 37

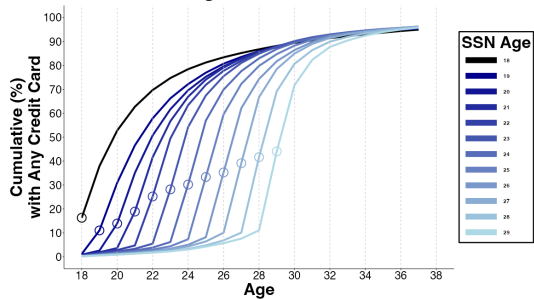
| Dep. Var: By Age 37, has ... | <u>Credit Card</u> |                   |                | <u>Auto Loan</u>    |                     |                    | <u>Mortgage</u>    |                    |                    |
|------------------------------|--------------------|-------------------|----------------|---------------------|---------------------|--------------------|--------------------|--------------------|--------------------|
|                              | (1)                | (2)               | (3)            | (4)                 | (5)                 | (6)                | (7)                | (8)                | (9)                |
| 21+                          | 0.93***<br>(0.19)  | 0.97***<br>(0.16) | 0.15<br>(0.16) | -10.24***<br>(0.38) | -10.22***<br>(0.37) | -6.87***<br>(0.46) | -2.19***<br>(0.55) | -2.28***<br>(0.56) | 0.39<br>(0.53)     |
| SSN Age                      | 0.03<br>(0.05)     | 0.10*<br>(0.04)   | 0.04<br>(0.03) | -1.48***<br>(0.10)  | -1.42***<br>(0.09)  | -1.44***<br>(0.10) | -1.34***<br>(0.1)  | -1.58***<br>(0.09) | -1.77***<br>(0.09) |
| Birth Year F.E.              |                    | X                 | X              |                     | X                   | X                  |                    | X                  | X                  |
| First Zip5 F.E.              |                    |                   | X              |                     |                     | X                  |                    |                    | X                  |
| $R^2$                        | < 0.001            | 0.002             | 0.025          | 0.008               | 0.009               | 0.039              | 0.002              | 0.005              | 0.063              |
| $N$                          | 6,122,932          | 6,122,932         | 6,122,932      | 6,122,932           | 6,122,932           | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932          |
| Mean, SSN Age <21            | 94.93              | 94.93             | 94.93          | 77.22               | 77.22               | 77.22              | 45.46              | 45.46              | 45.46              |

# Credit Access By Age 37 With Additional F.E.

| Dep Var: By Age 37, has... | <u>Credit Card</u> |           |           | <u>Auto Loan</u> |           |           | <u>Mortgage</u> |           |           |
|----------------------------|--------------------|-----------|-----------|------------------|-----------|-----------|-----------------|-----------|-----------|
|                            | (1)                | (2)       | (3)       | (4)              | (5)       | (6)       | (7)             | (8)       | (9)       |
| 21+                        | -0.02              | -0.05     | 0.50***   | -6.18***         | -6.16***  | -4.43***  | 1.09*           | 1.13*     | 2.33***   |
|                            | (0.12)             | (0.12)    | (0.14)    | (0.34)           | (0.35)    | (0.20)    | (0.49)          | (0.46)    | (0.35)    |
| SSN Age                    | 0.02               | 0.02      | 0.15***   | -1.48***         | -1.47***  | -1.06***  | -1.86***        | -1.85***  | -1.56***  |
|                            | (0.03)             | (0.03)    | (0.04)    | (0.09)           | (0.09)    | (0.11)    | (0.08)          | (0.08)    | (0.09)    |
| Birth Year F.E.            | X                  | X         | X         | X                | X         | X         | X               | X         | X         |
| First Zip5 F.E.            | X                  | X         | X         | X                | X         | X         | X               | X         | X         |
| Last Zip5 F.E.             | X                  | X         | X         | X                | X         | X         | X               | X         | X         |
| Longest Zip5 F.E.          |                    | X         | X         |                  | X         | X         |                 | X         | X         |
| Number Zip5 F.E.           |                    |           | X         |                  |           | X         |                 |           | X         |
| $R^2$                      | 0.040              | 0.042     | 0.060     | 0.076            | 0.079     | 0.124     | 0.117           | 0.127     | 0.146     |
| $N$                        | 6,122,932          | 6,122,932 | 6,122,932 | 6,122,932        | 6,122,932 | 6,122,932 | 6,122,932       | 6,122,932 | 6,122,932 |
| Mean, SSN Age <21          | 94.93              | 94.93     | 94.93     | 77.22            | 77.22     | 77.22     | 45.46           | 45.46     | 45.46     |

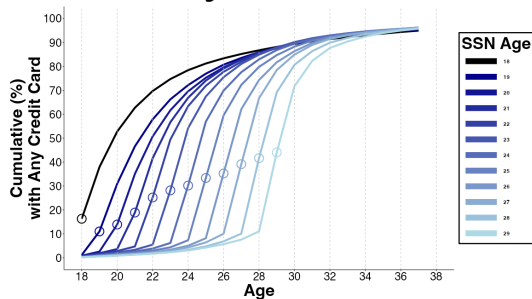
# Immigrants' Access to Credit Cards Converges With Non-Immigrants

## A. Any Credit Card

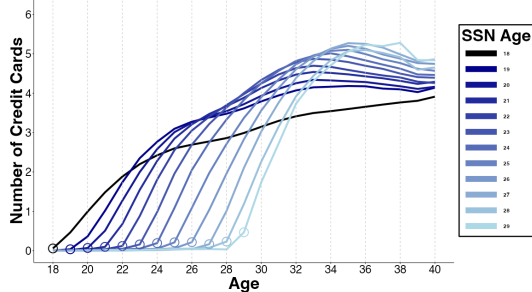


# Immigrants' Access to Credit Cards Converges With Non-Immigrants

## A. Any Credit Card

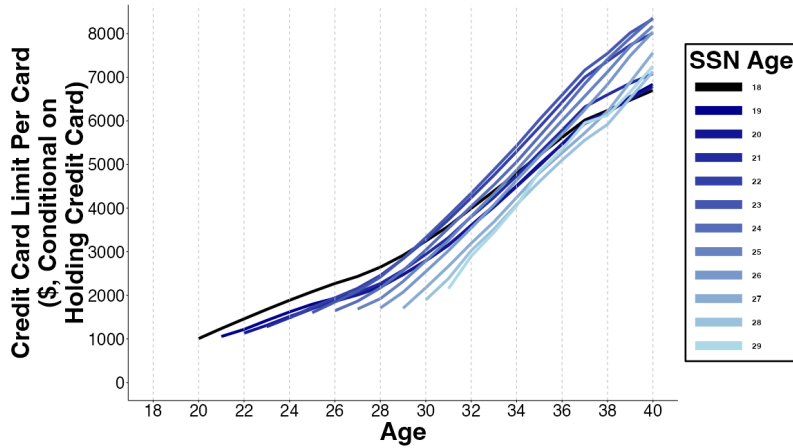


## B. Number of Credit Cards



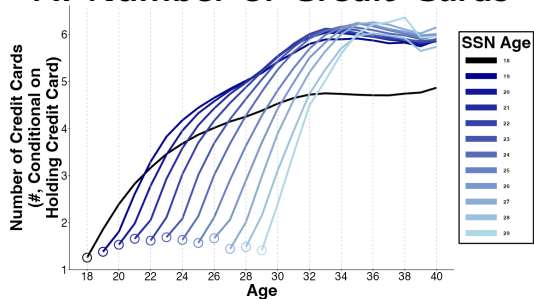


# Credit Card Limit Per Card 🚀



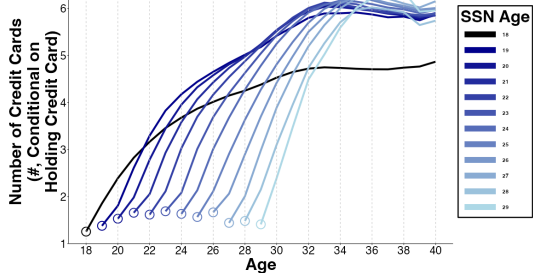
# Immigrants' Credit Card Access, Conditional On Holding Any Credit Card

## A. Number of Credit Cards

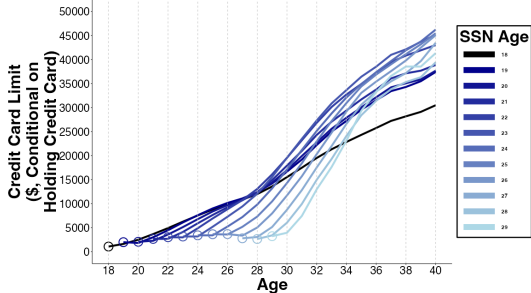


# Immigrants' Credit Card Access, Conditional On Holding Any Credit Card

## A. Number of Credit Cards



## B. Credit Card Limits



# Number of Credit Cards At Age 30 📌

|                   | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                |
|-------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| 21+               | 1.52***<br>(0.07)  | 1.51***<br>(0.07)  | 1.25***<br>(0.08)  | 1.18***<br>(0.06)  | 1.17***<br>(0.06)  | 1.29***<br>(0.05)  |
| SSN Age           | -0.29***<br>(0.01) | -0.31***<br>(0.01) | -0.32***<br>(0.01) | -0.33***<br>(0.01) | -0.33***<br>(0.01) | -0.30***<br>(0.01) |
| F.E. Birth Year   |                    | X                  | X                  | X                  | X                  | X                  |
| F.E. First Zip5   |                    |                    | X                  | X                  | X                  | X                  |
| F.E. Last Zip5    |                    |                    |                    | X                  | X                  | X                  |
| F.E. Longest Zip5 |                    |                    |                    |                    | X                  | X                  |
| F.E. Number Zip5  |                    |                    |                    |                    |                    | X                  |
| $R^2$             | 0.002              | 0.007              | 0.058              | 0.091              | 0.097              | 0.123              |
| $N$               | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932          |
| Mean, SSN Age <21 | 3.42               | 3.42               | 3.42               | 3.42               | 3.42               | 3.42               |

## Number of Credit Cards At Age 40 🐦

|                   | (1)               | (2)               | (3)               | (4)               | (5)               | (6)               |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 21+               | 0.33***<br>(0.04) | 0.33***<br>(0.04) | 0.01<br>(0.04)    | -0.01<br>(0.04)   | -0.02<br>(0.04)   | 0.17***<br>(0.04) |
| SSN Age           | 0.10***<br>(0.01) | 0.10***<br>(0.01) | 0.09***<br>(0.01) | 0.08***<br>(0.01) | 0.08***<br>(0.01) | 0.11***<br>(0.01) |
| F.E. Birth Year   |                   | X                 | X                 | X                 | X                 | X                 |
| F.E. First Zip5   |                   |                   | X                 | X                 | X                 | X                 |
| F.E. Last Zip5    |                   |                   |                   | X                 | X                 | X                 |
| F.E. Longest Zip5 |                   |                   |                   |                   | X                 | X                 |
| F.E. Number Zip5  |                   |                   |                   |                   |                   | X                 |
| $R^2$             | 0.002             | 0.003             | 0.028             | 0.058             | 0.060             | 0.092             |
| $N$               | 4,800,195         | 4,800,195         | 4,800,195         | 4,800,195         | 4,800,195         | 4,800,195         |
| Mean, SSN Age <21 | 3.97              | 3.97              | 3.97              | 3.97              | 3.97              | 3.97              |

## Credit Card Limits At Age 30 📍

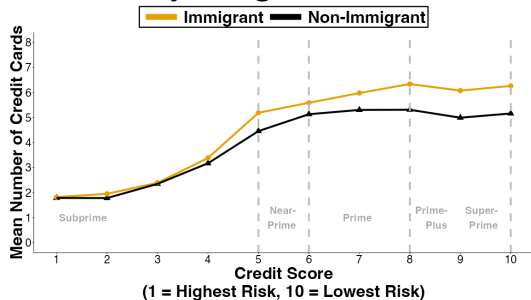
|                   | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|-------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| 21+               | 4982.1***<br>(528.6) | 4948.6***<br>(571.3) | 3788.2***<br>(489.8) | 3307.3***<br>(406.9) | 3248.8***<br>(386.8) | 3656.3***<br>(373.2) |
| SSN Age           | -1288.4***<br>(79.9) | -1360.7***<br>(77.2) | -1490.8***<br>(74.2) | -1545.1***<br>(68.5) | -1542.4***<br>(68.7) | -1446.1***<br>(67.2) |
| F.E. Birth Year   |                      | X                    | X                    | X                    | X                    | X                    |
| F.E. First Zip5   |                      |                      | X                    | X                    | X                    | X                    |
| F.E. Last Zip5    |                      |                      |                      | X                    | X                    | X                    |
| F.E. Longest Zip5 |                      |                      |                      |                      | X                    | X                    |
| F.E. Number Zip5  |                      |                      |                      |                      |                      | X                    |
| $R^2$             | 0.002                | 0.009                | 0.080                | 0.124                | 0.130                | 0.142                |
| $N$               | 6,122,932            | 6,122,932            | 6,122,932            | 6,122,932            | 6,122,932            | 6,122,932            |
| Mean, SSN Age <21 | 11,733.1             | 11,733.1             | 11,733.1             | 11,733.1             | 11,733.1             | 11,733.1             |

## Credit Card Limits At Age 40 📍

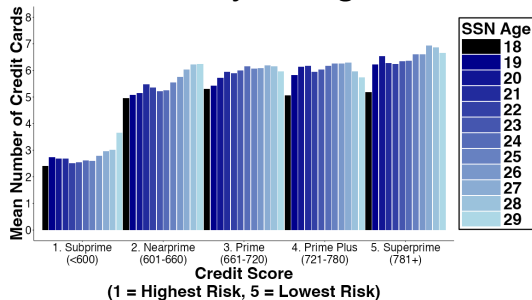
|                   | (1)                  | (2)                  | (3)                | (4)              | (5)              | (6)                  |
|-------------------|----------------------|----------------------|--------------------|------------------|------------------|----------------------|
| 21+               | 4262.0***<br>(744.4) | 4327.3***<br>(712.9) | 1633.6*<br>(536.3) | 639.6<br>(353.4) | 526.8<br>(327.8) | 1696.6***<br>(385.8) |
| SSN Age           | 318.3<br>(188.7)     | 456.9*<br>(149.1)    | 192.4<br>(135.4)   | 59.7<br>(110.3)  | 61.5<br>(111.7)  | 273.0<br>(137.9)     |
| F.E. Birth Year   |                      | X                    | X                  | X                | X                | X                    |
| F.E. First Zip5   |                      |                      | X                  | X                | X                | X                    |
| F.E. Last Zip5    |                      |                      |                    | X                | X                | X                    |
| F.E. Longest Zip5 |                      |                      |                    |                  | X                | X                    |
| F.E. Number Zip5  |                      |                      |                    |                  |                  | X                    |
| $R^2$             | 0.002                | 0.008                | 0.082              | 0.151            | 0.157            | 0.181                |
| $N$               | 4,800,195            | 4,800,195            | 4,800,195          | 4,800,195        | 4,800,195        | 4,800,195            |
| Mean, SSN Age <21 | 22,477.2             | 22,477.2             | 22,477.2           | 22,477.2         | 22,477.2         | 22,477.2             |

# Number Of Credit Cards At Age 37, Conditional On Age 30 Credit Score 📍

## A. By Immigration Status



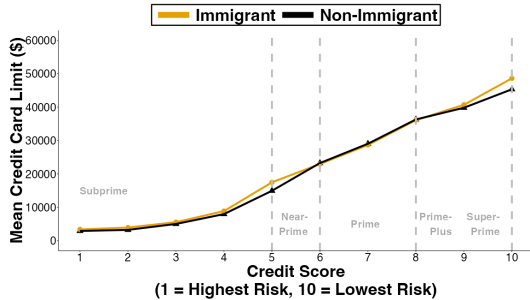
## B. By SSN Age



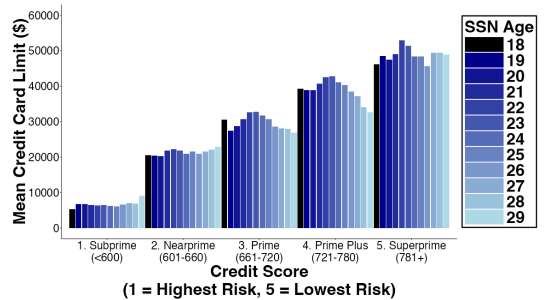


# Credit Card Limits At Age 37, Conditional On Age 30 Credit Score 🚀

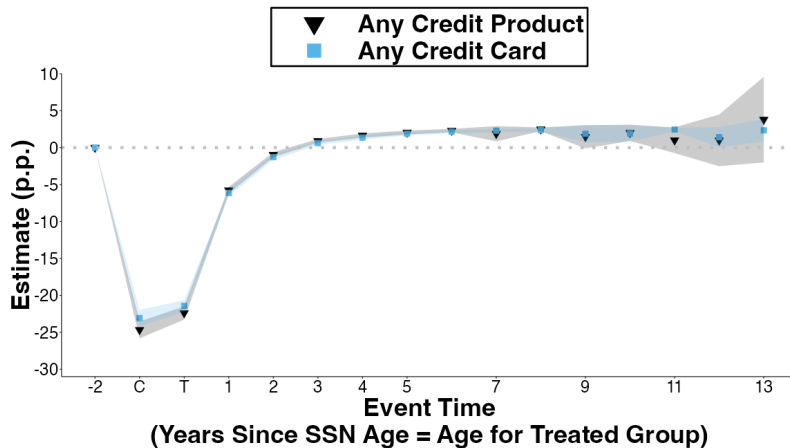
## A. By Immigration Status



## B. By SSN Age

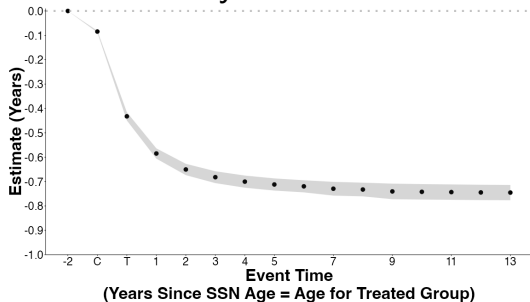


## Paired Cohorts: First Credit Product & First Credit Card 🚀

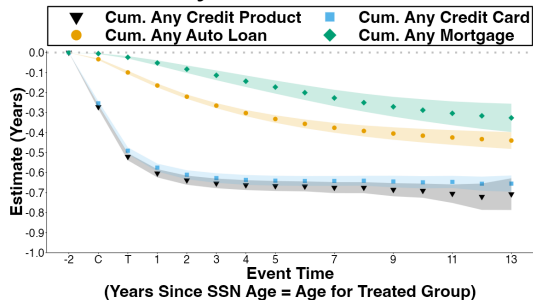


# Paired Cohorts: Cumulatives 🚀

## A. Any Credit Score



## B. By Credit Product

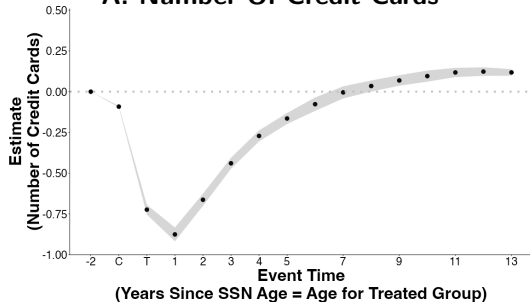


## Paired Cohorts: Cumulative Years of Delay 🚀

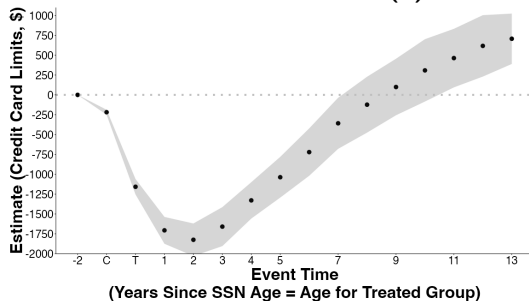
|                         | <u>Any Credit Score</u> | <u>Credit Product</u> | <u>Credit Card</u>   | <u>Auto Loan</u>     | <u>Mortgage</u>      |
|-------------------------|-------------------------|-----------------------|----------------------|----------------------|----------------------|
|                         | (1)                     | (2)                   | (3)                  | (4)                  | (5)                  |
| C                       | -0.085***<br>(0.002)    | -0.272***<br>(0.005)  | -0.254***<br>(0.005) | -0.033***<br>(0.002) | -0.005***<br>(0.001) |
| T                       | -0.432***<br>(0.009)    | -0.521***<br>(0.009)  | -0.491***<br>(0.009) | -0.100***<br>(0.004) | -0.024***<br>(0.003) |
| 5                       | -0.712***<br>(0.013)    | -0.667***<br>(0.013)  | -0.640***<br>(0.013) | -0.332***<br>(0.010) | -0.173***<br>(0.017) |
| 10                      | -0.742***<br>(0.016)    | -0.690***<br>(0.018)  | -0.648***<br>(0.016) | -0.415***<br>(0.017) | -0.288***<br>(0.028) |
| <i>N</i> Paired Cohorts | 68                      | 68                    | 68                   | 68                   | 68                   |
| <i>N</i> Consumers      | 403,506                 | 403,506               | 403,506              | 403,506              | 403,506              |
| <i>N</i>                | 6,456,096               | 6,456,096             | 6,456,096            | 6,456,096            | 6,456,096            |

# Paired Cohorts: Credit Cards 🚀

## A. Number Of Credit Cards



## B. Credit Card Limits (\$)



## Potential Mechanism #2: Emigration

Proxy emigration by **attrition** from data

Immigrants exit data more than Non-Immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

## Potential Mechanism #2: Emigration

Proxy emigration by **attrition** from data

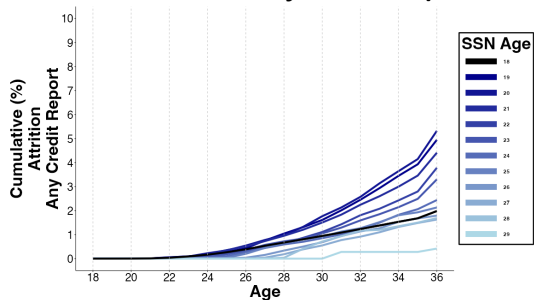
Immigrants exit data more than Non-Immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

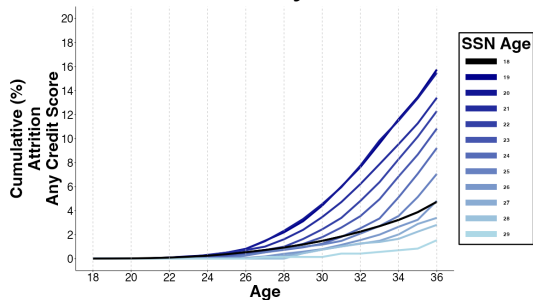
Restrict to consumers observed in 2024:

- **Mortgage** Immigrant vs. Non-Immigrant gap closes but SSN Age gap remains
- **Auto loan** Immigrant vs. Non-Immigrant gap & SSN Age gap remains
- **Credit card** dynamics by SSN Age remains

## A. Attrition In Any Credit Report



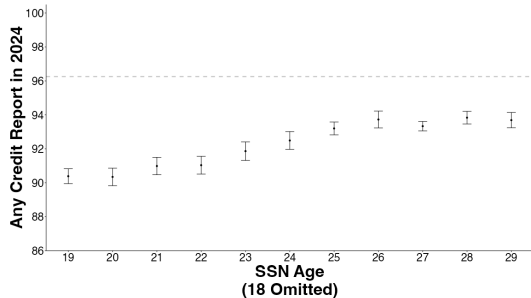
## B. Attrition In Any Credit Score



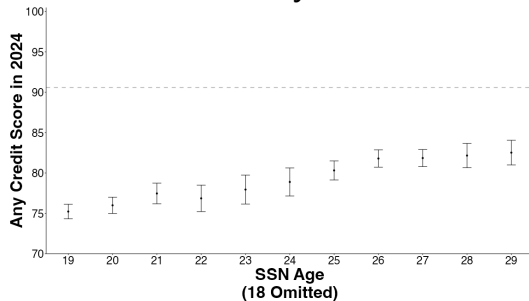


# Consumers Observed In 2024 🚀

## A. Attrition In Any Credit Report

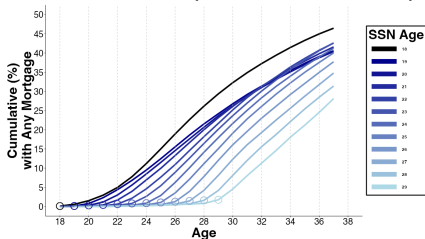


## B. Attrition In Any Credit Score

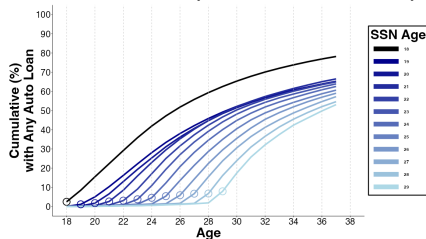


# Lifecycle Of Credit For Consumers Observed In 2024 📍

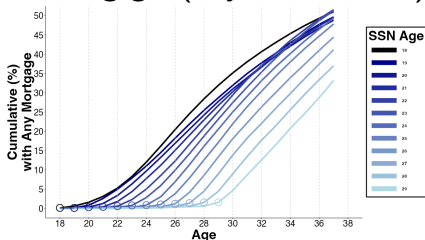
## A. Mortgages (Any 2024 Report)



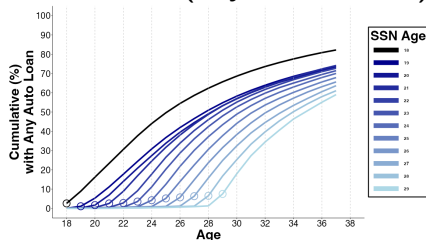
## B. Auto Loan (Any 2024 Report)



## C. Mortgages (Any 2024 Product)

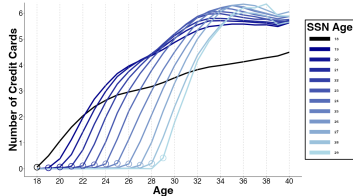


## D. Auto Loan (Any 2024 Product)

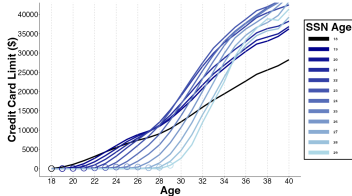


# Lifecycle Of Credit For Consumers Observed In 2024 (Any Tradeline) 🚀

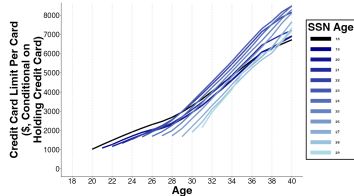
## A. Number Credit Cards



## B. Credit Card Limits



## C. Credit Limit Per Card



# Consumers Observed In Data In 2024 With Additional F.E. 🚀

| Dep Var: In 2024, has... | Any Credit Report      |                        | Any Credit Score       |                        | Any Credit Tradeline   |                        |
|--------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                          | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    |
| 21+                      | -0.0543***<br>(0.0020) | -0.0485***<br>(0.0022) | -0.1404***<br>(0.0070) | -0.1216***<br>(0.0048) | -0.1442***<br>(0.0063) | -0.1207***<br>(0.0042) |
| SSN Age                  | 0.0043***<br>(0.0005)  | 0.0052***<br>(0.0004)  | 0.0080***<br>(0.0009)  | 0.0109***<br>(0.0007)  | 0.0068***<br>(0.0007)  | 0.0108***<br>(0.0006)  |
| Birth Year F.E.          | X                      | X                      | X                      | X                      | X                      | X                      |
| First Zip5 F.E.          | X                      | X                      | X                      | X                      | X                      | X                      |
| Last Zip5 F.E.           |                        | X                      |                        | X                      |                        | X                      |
| Longest Zip5 F.E.        |                        | X                      |                        | X                      |                        | X                      |
| Number Zip5 F.E.         |                        | X                      |                        | X                      |                        | X                      |
| $R^2$                    | 0.007                  | 0.034                  | 0.020                  | 0.111                  | 0.023                  | 0.138                  |
| $N$                      | 6,122,932              | 6,122,932              | 6,122,932              | 6,122,932              | 6,122,932              | 6,122,932              |
| Mean, SSN Age <21        | 0.9615                 | 0.9615                 | 0.9031                 | 0.9031                 | 0.8503                 | 0.8503                 |

# Credit Access By Age 37 With Additional F.E. Conditional On Any Credit Score Observed in 2024 📍

| Dep Var: By Age 37, has... | <u>Credit Card</u> |                   |                   | <u>Auto Loan</u>   |                    |                    | <u>Mortgage</u>    |                    |                    |
|----------------------------|--------------------|-------------------|-------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                            | (1)                | (2)               | (3)               | (4)                | (5)                | (6)                | (7)                | (8)                | (9)                |
| 21+                        | 0.56***<br>(0.11)  | 0.51***<br>(0.11) | 0.67***<br>(0.13) | -2.56***<br>(0.34) | -2.56***<br>(0.35) | -2.01***<br>(0.22) | 6.33***<br>(0.49)  | 6.31***<br>(0.48)  | 6.57***<br>(0.41)  |
| SSN Age                    | -0.04<br>(0.03)    | -0.03<br>(0.03)   | 0.08<br>(0.04)    | -1.83***<br>(0.10) | -1.83***<br>(0.10) | -1.46***<br>(0.12) | -2.45***<br>(0.08) | -2.44***<br>(0.08) | -2.20***<br>(0.11) |
| Birth Year F.E.            | X                  | X                 | X                 | X                  | X                  | X                  | X                  | X                  | X                  |
| First Zip5 F.E.            | X                  | X                 | X                 | X                  | X                  | X                  | X                  | X                  | X                  |
| Last Zip5 F.E.             | X                  | X                 | X                 | X                  | X                  | X                  | X                  | X                  | X                  |
| Longest Zip5 F.E.          |                    | X                 | X                 |                    | X                  | X                  |                    | X                  | X                  |
| Number Zip5 F.E.           |                    |                   | X                 |                    |                    | X                  |                    |                    | X                  |
| $R^2$                      | 0.038              | 0.042             | 0.054             | 0.067              | 0.071              | 0.099              | 0.114              | 0.125              | 0.134              |
| $N$                        | 5,489,726          | 5,489,726         | 5,489,726         | 5,489,726          | 5,489,726          | 5,489,726          | 5,489,726          | 5,489,726          | 5,489,726          |
| Mean, SSN Age <21          | 96.05              | 96.05             | 96.05             | 80.37              | 80.37              | 80.37              | 48.76              | 48.76              | 48.76              |

# Use Survey of Consumer Finances (SCF) 2022 To Evaluate Mechanisms

SCF 2022 includes if Immigrant & Age of Immigration

Adapt earlier cross-sectional regression to this data:

$$Y_i = \beta_1 \cdot \mathbb{I}\{\text{Immigration Age } 21+\}_i + \beta_2 \cdot \text{Immigration Age}_i + \mu_{b(i)} + \eta_{d(i)} + \epsilon_i$$

$\eta_{d(i)}$  are F.E. Demographic Controls: male, race, hispanic, spouse, male spouse, marital status, household size, number of children, education levels, & quintiles of household wage income

# Use Survey of Consumer Finances (SCF) 2022 To Evaluate Mechanisms

SCF 2022 includes if Immigrant & Age of Immigration

Adapt earlier cross-sectional regression to this data:

$$Y_i = \beta_1 \cdot \mathbb{I}\{\text{Immigration Age } 21+\}_i + \beta_2 \cdot \text{Immigration Age}_i + \mu_{b(i)} + \eta_{d(i)} + \epsilon_i$$

$\eta_{d(i)}$  are F.E. Demographic Controls: male, race, hispanic, spouse, male spouse, marital status, household size, number of children, education levels, & quintiles of household wage income

## SCF Debt Attitude Questions:

*“People have many different reasons for borrowing money which they pay back over a period of time. For each of the reasons I read, please tell me whether you feel it is all right for someone like yourself to borrow money...to finance the purchase of a car.”*

# Immigrants More Likely To Buy Cars With Cash (SCF, 2022)

12.5 p.p. [95% C.I. -2.2, 27.3], 17% Vs. Non-Immigrants

| Dep Var:                  | <u>Conditional On</u><br><u>Having A Car</u> |                       | <u>Conditional On</u><br><u>Homeowner</u> |                      |
|---------------------------|----------------------------------------------|-----------------------|-------------------------------------------|----------------------|
|                           | <u>Has Car</u>                               | <u>Auto Cash</u>      | <u>Homeowner</u>                          | <u>No Mortgage</u>   |
|                           | (1)                                          | (2)                   | (3)                                       | (4)                  |
| 21+                       | -0.0620<br>(0.0477)                          | 0.1254*<br>(0.0753)   | 0.0241<br>(0.0675)                        | 0.0704<br>(0.0791)   |
| Immigration Age           | -0.0003<br>(0.0044)                          | -0.0164**<br>(0.0077) | -0.0155***<br>(0.0055)                    | -0.0100*<br>(0.0058) |
| F.E. Birth Year           | X                                            | X                     | X                                         | X                    |
| F.E. Demographic Controls | X                                            | X                     | X                                         | X                    |
| $R^2$                     | 0.156                                        | 0.038                 | 0.298                                     | 0.175                |
| $N$                       | 1,813                                        | 1,568                 | 1,813                                     | 883                  |
| Mean, Immigration Age <21 | 0.8923                                       | 0.7396                | 0.5398                                    | 0.1619               |

\*\*\* 1%, \*\* , 5%, \* 10% statistical significance



# Immigrants Have ↓ Demand for Debt (↑ Auto, ↑ Credit Cards, ↑ Mortgage. All Attenuated By Immigration Age) To Non-Immigrants (SCF, 2022) ...But Low Statistical Power To Detect Out Differences

| Dep Var: Apply For...     | <u>Any Credit</u><br>(1) | <u>Auto</u><br>(2)  | <u>Mortgage</u><br>(3) | <u>Refinance Mortgage</u><br>(4) | <u>Credit Card</u><br>(5) | <u>Limit Increase</u><br>(6) | <u>Other Credit</u><br>(7) |
|---------------------------|--------------------------|---------------------|------------------------|----------------------------------|---------------------------|------------------------------|----------------------------|
| 21+                       | -0.0071<br>(0.0695)      | -0.0690<br>(0.0739) | 0.0625<br>(0.0700)     | 0.0437<br>(0.0401)               | 0.0519<br>(0.0843)        | -0.0279<br>(0.0609)          | -0.0729<br>(0.0458)        |
| Immigration Age           | 0.0050<br>(0.0060)       | 0.0112<br>(0.0072)  | -0.0036<br>(0.0067)    | -0.0048<br>(0.0039)              | -0.0102<br>(0.0086)       | 0.0073<br>(0.0062)           | 0.0128**<br>(0.0055)       |
| F.E. Birth Year           | X                        | X                   | X                      | X                                | X                         | X                            | X                          |
| F.E. Demographic Controls | X                        | X                   | X                      | X                                | X                         | X                            | X                          |
| $R^2$                     | 0.059                    | 0.059               | 0.056                  | 0.033                            | 0.039                     | 0.043                        | 0.040                      |
| $N$                       | 1,813                    | 1,813               | 1,813                  | 1,813                            | 1,813                     | 1,813                        | 1,813                      |
| Mean, Immigration Age <21 | 0.6481                   | 0.1995              | 0.1128                 | 0.0670                           | 0.4043                    | 0.1533                       | 0.1208                     |

\*\*\* 1%, \*\*, 5%, \* 10% statistical significance

# Immigrants Have ↓ Demand for Debt (↑ Auto, ↑ Credit Cards, ↑ Mortgage. All Attenuated By Immigration Age) To Non-Immigrants (SCF, 2022) ...But Low Statistical Power To Detect Out Differences

| Dep Var: Apply For...     | <u>Any Credit</u><br>(1) | <u>Auto</u><br>(2)  | <u>Mortgage</u><br>(3) | <u>Refinance Mortgage</u><br>(4) | <u>Credit Card</u><br>(5) | <u>Limit Increase</u><br>(6) | <u>Other Credit</u><br>(7) |
|---------------------------|--------------------------|---------------------|------------------------|----------------------------------|---------------------------|------------------------------|----------------------------|
| 21+                       | -0.0071<br>(0.0695)      | -0.0690<br>(0.0739) | 0.0625<br>(0.0700)     | 0.0437<br>(0.0401)               | 0.0519<br>(0.0843)        | -0.0279<br>(0.0609)          | -0.0729<br>(0.0458)        |
| Immigration Age           | 0.0050<br>(0.0060)       | 0.0112<br>(0.0072)  | -0.0036<br>(0.0067)    | -0.0048<br>(0.0039)              | -0.0102<br>(0.0086)       | 0.0073<br>(0.0062)           | 0.0128**<br>(0.0055)       |
| F.E. Birth Year           | X                        | X                   | X                      | X                                | X                         | X                            | X                          |
| F.E. Demographic Controls | X                        | X                   | X                      | X                                | X                         | X                            | X                          |
| $R^2$                     | 0.059                    | 0.059               | 0.056                  | 0.033                            | 0.039                     | 0.043                        | 0.040                      |
| $N$                       | 1,813                    | 1,813               | 1,813                  | 1,813                            | 1,813                     | 1,813                        | 1,813                      |
| Mean, Immigration Age <21 | 0.6481                   | 0.1995              | 0.1128                 | 0.0670                           | 0.4043                    | 0.1533                       | 0.1208                     |

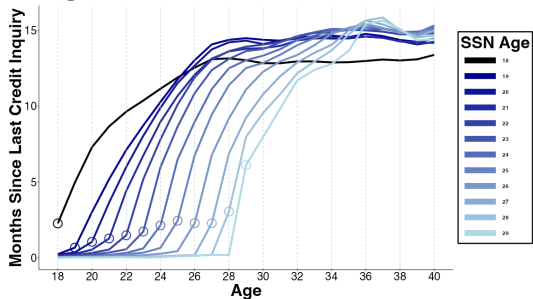
\*\*\* 1%, \*\*, 5%, \* 10% statistical significance

**Credit inquiries data in credit panel shows**  
 ↑ immigrant credit demand in 20s, ↓ demand in 30s

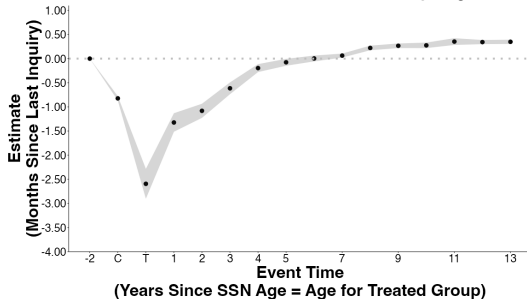
# Does Lower Credit Demand Explain Lower Access?

Immigrants Have  $\uparrow$  Credit Demand In 20s,  $\downarrow$  Demand In 30s 🚀

**A. Months Since Last Credit Inquiry**  
Higher Value = Lower Credit Demand



**B. Paired Cohorts**  
Months Since Last Credit Inquiry



**Caveat:** Lower credit demand in 30s may be function of earlier rejections  $\rightarrow$  deterrence

# Inconclusive Evidence On Other Demand and Supply Proxies (SCF, 2022)

## ...Given Low Statistical Power To Detect Out Differences

| (Conditional On...<br>Dep Var: | Why Did Not Apply For Credit? |                     | Outcome After Applying For Credit? |                     |                      |
|--------------------------------|-------------------------------|---------------------|------------------------------------|---------------------|----------------------|
|                                | ...Not Applying)              |                     | ...Applying)                       | ...Reject)          | ...Reject & Reapply) |
|                                | No Demand                     | Rate Too High       | Reject/Less                        | Reapply             | Reapply Reject       |
|                                | (1)                           | (2)                 | (3)                                | (4)                 | (5)                  |
| 21+                            | 0.0862<br>(0.1261)            | -0.0015<br>(0.0827) | -0.0156<br>(0.0734)                | -0.1510<br>(0.2250) | 0.4863**<br>(0.1917) |
| Immigration Age                | -0.0077<br>(0.0111)           | 0.0121<br>(0.0088)  | -0.0060<br>(0.0061)                | 0.0270<br>(0.0188)  | -0.0089<br>(0.0204)  |
| F.E. Birth Year                | X                             | X                   | X                                  | X                   | X                    |
| F.E. Demographic Controls      | X                             | X                   | X                                  | X                   | X                    |
| $R^2$                          | 0.177                         | 0.095               | 0.085                              | 0.092               | 0.340                |
| $N$                            | 666                           | 666                 | 1,147                              | 292                 | 178                  |
| Mean, Immigration Age <21      | 0.7952                        | 0.0753              | 0.2707                             | 0.603               | 0.6022               |

\*\*\* 1%, \*\* , 5%, \* 10% statistical significance